Cross-Sectional Momentum Across Market Regimes – A Machine Learning Approach

By

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- **Proofreading and clarity.** Grammar correction, copy-editing, and suggestions for improving the readability of the author's drafts.
- **Idea generation.** Brainstorming structural framings, exploring alternative phrasings, and surfacing candidate literature.
- **Code support.** Code review, refactoring suggestions, and debugging assistance for the analytical scripts.
- **Verification.** Cross-checking consistency between prose, tables, and source data.

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Management Summary

Cross-sectional momentum, the tendency for recent stock-market winners to keep winning and losers to keep losing, is one of the most profitable anomalies in finance and one of the most fragile. Momentum portfolios suffered losses exceeding 50% during the 2009 market rebound and crash repeatedly around regime transitions. This thesis examines what form cross-sectional momentum's regime-dependence takes, and how a nonlinear model captures and exploits that form. The aim is to understand the mechanism, not to propose a strategy that outperforms the market on aggregate.

Introduction (Section 1)

The introduction sets out the empirical setting (U.S. equities 1990–2025, with UK and Japan replication) and defines cross-sectional momentum, then motivates the study through the anomaly's fragility around regime transitions and identifies the gap in prior crash-management work, which treats the underlying selection rule as unconditional and adjusts only its exposure or horizon weighting. It poses the research question, how the cross-section of momentum reorganises across regimes and what mechanism drives that change, and outlines an approach that combines a Bayesian HMM regime signal with the momentum term structure inside a nonlinear cross-sectional model, whose selections are then opened up to characterise the mechanism. The section closes with the headline findings, the two contributions (the methodological combination and the mechanism it reveals), and an outline of the remaining sections.

Literature (Section 2)

The thesis sits at the intersection of four literatures. Standard cross-sectional momentum at intermediate horizons (Jegadeesh and Titman, 1993; Novy-Marx, 2012), with behavioural drivers in gradual information diffusion (Hong and Stein, 1999); momentum crashes and exposure-based risk management, where Barroso and Santa-Clara (2015) and others reduce gross exposure during stress; regime-conditional asset pricing, where Cooper et al. (2004) and Goulding et al. (2023) use binary or weight-blending regime treatments; and machine learning in cross-sectional return prediction (Gu et al., 2020). The thesis investigates how these strands interact, characterising the mechanism by which the cross-sectional structure of momentum reorganises across regimes when a continuous probabilistic regime signal and the full momentum term structure are jointly fed to a nonlinear cross-sectional model.

Methodology (Section 3)

A two-stage framework. Stage one estimates a Bayesian two-state Hidden Markov Model via Gibbs sampling on four monthly stress indicators, producing a filtered probability of market panic. Stage two combines this regime signal with twelve momentum horizons in models of increasing complexity (a deterministic formula, a logistic regression, and an

XGBoost ensemble), evaluated in long-short construction with NYSE breakpoints and value weighting.[1]

Data (Section 4)

The empirical analysis spans 35 years of U.S. equities from CRSP (1990–2025), split into a 1990–2010 training window and a 167-month 2011–2025 out-of-sample test, with delisting returns retained to avoid survivorship bias. Rather than picking the HMM's input features by hand, a four-pass selection pipeline filters candidates through a quality screen, ranks them by portfolio Sharpe, validates the leader at higher seed counts, and confirms its stability across seeds. The framework is then taken outside the U.S., with the entire pipeline rerun on Compustat Global data for the United Kingdom and Japan and the HMM features re-selected per region, providing an external check that the mechanism is not a U.S. artefact.

Results (Section 5)

The headline finding is that the cross-section of momentum reorganises systematically across regimes, and capturing this requires both a probabilistic regime signal and a nonlinear model. XGBoost on the regime signal and the momentum term structure delivers a six-factor alpha of 24.1% ($t = 4.81$) and an out-of-sample Sharpe of 1.11, with the advantage concentrated in panic (1.53 versus 0.84 in calm). A linear baseline on identical features collapses, and removing the regime signal alone cuts the Sharpe by more than 60%. The novel contribution lies in examining the model's selections: cluster analysis of the long leg's cross-sectional z-curves recovers four recurring selection modes that trace a single market cycle, and the model avoids the leg-beta inversion that drives traditional momentum's crashes in every regime.

Conclusion (Section 6)

Cross-sectional momentum's regime-dependent structure is real and characterisable: the regime signal acts as a *context variable*, conditioning how a nonlinear cross-sectional model uses the momentum term structure rather than predicting returns directly. The thesis makes two contributions: the methodological combination of a continuous probabilistic regime signal with a nonlinear cross-sectional model is itself novel, and examining the model reveals a mechanism in which the selection direction inverts in panic while the portfolio stays fully invested, in contrast to the exposure-scaling responses in the prior literature. The primary limitation is reversal-failure vulnerability in prolonged fundamental deterioration; the most direct future extension is a hybrid that layers an exposure-scaling overlay on top of XGB's stock-selection.

[1] NYSE breakpoints construct decile cutoffs from NYSE-listed stocks only and apply them across the full NYSE/AMEX/NASDAQ universe, preventing the small-cap-heavy AMEX/NASDAQ tails from dominating the extreme deciles; value weighting assigns each stock a portfolio weight proportional to its market capitalisation. Both are standard conventions in the cross-sectional asset-pricing literature.

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Appendices

1 Introduction

This thesis applies machine learning to study the behaviour of cross-sectional momentum[2] across market regimes. The setting is the U.S. equity cross-section from January 1990 to November 2025, with international replication in the United Kingdom and Japan. The central object of study is the cross-sectional momentum anomaly, the tendency for recent stock-market winners to continue outperforming and losers to continue underperforming (Jegadeesh and Titman, 1993). The aim is to better understand this anomaly conditional on the market regime, not to propose a strategy that outperforms the market on aggregate.

Cross-sectional momentum is one of the most robust anomalies in empirical finance and one of the most fragile. Momentum portfolios suffered losses exceeding 50% during the 2009 market rebound, and similar crashes have recurred around regime transitions. This fragility in the method motivates a substantial literature on momentum-crash management, but existing responses share a structural assumption. The regime signal enters only to scale or re-weight a fixed scoring rule, never as a feature inside the rule itself. Barroso and Santa-Clara (2015) targets constant volatility during stress, while Cooper et al. (2004) treat regimes as a binary up-or-down variable and Goulding et al. (2023) blend fast and slow horizon weights. What this leaves unexamined is how the full momentum term structure and the regime signal jointly play out, that is, the mechanism by which the cross-sectional structure of momentum reorganises across regimes.

This thesis examines that possibility. The regime signal acts as a *context variable*: it carries no direct return-predictive content, but conditions how a nonlinear cross-sectional model uses the momentum term structure in a given market state. No prior study has documented how the full term structure of cross-sectional momentum reorganises across regimes, or tested whether capturing this reorganisation requires nonlinear modelling.

Research question

The thesis asks how the cross-sectional structure of momentum changes between calm and panic regimes, and what mechanism drives that change.

Approach

The question is answered in three steps. First, a Bayesian Hidden Markov Model is estimated on four monthly macrofinancial stress indicators (drawdown, return dispersion, relative market participation, and credit spread), producing a filtered probability of market panic that updates monthly. This regime signal is combined with the twelve momentum horizons in cross-sectional stock-selection models (deterministic formula, logistic regression, XGBoost ensemble), each evaluated out-of-sample on 2011–2025 U.S.

[2] Cross-sectional momentum ranks stocks against each other at each point in time, going long the relative winners and short-selling the relative losers within the same universe. This contrasts with time-series momentum, which trades each asset based on its own past return regardless of the cross-section.

equities in long-short construction. Second, the chosen model's selection rule is opened up month by month with SHAP decomposition and a clustering of long-leg cross-sectional z-curves, identifying the role the regime signal plays in selection. Third, the strategy is tested outside the main test sample, on a 25-year expanding-window backtest covering the dot-com bust, the GFC, and the 2009 momentum crash, and on UK and Japanese equities with locally-fitted HMMs.

Headline findings

On 2011–2025 U.S. equities, the XGBoost ensemble on π_t^{filter} and the twelve momentum horizons produces a six-factor alpha of 24.1% ($t = 4.81$), while a linear baseline on identical features collapses and removing π_t^{filter} alone cuts XGBoost's Sharpe by more than 60%. The mechanism is that the cross-section of momentum reorganises systematically across regimes: in panic, the regime signal flips the model's selection direction while keeping the long-short portfolio fully invested, the long leg holds stocks not top-ranked at any individual momentum horizon, and the leg-beta inversion that drives traditional momentum's crashes is absent in every regime. Cluster analysis recovers four recurring selection modes that trace a single market cycle.

Contributions

The thesis makes two contributions to the momentum literature. First, the methodological combination of a continuous probabilistic regime filter (the Bayesian HMM filtered probability π_t^{filter}) with a nonlinear cross-sectional model (XGBoost) is itself novel relative to the linear and rank-based regime-conditional rules in the prior literature. Second, examining the model's selections cluster by cluster (Section 5.2.3) reveals a regime-conditional mechanism in which the cross-sectional structure of momentum reorganises across regimes, in contrast to the exposure-scaling, horizon-shifting, and unconditional signal-reweighting responses of the existing crash-management literature.

Outline

The remainder of the thesis is organised as follows. Section 2 (Literature Review) reviews the relevant literature on momentum's horizon structure, momentum crashes and exposure-based risk management, regime models in asset pricing, and machine learning in cross-sectional finance. Section 3 (Methodology) develops the two-stage framework, including the Bayesian HMM specification and Gibbs sampling estimation, and the cross-sectional stock-selection methods. Section 4 (Data) describes the U.S. equity sample, the four-pass HMM feature selection pipeline, and the cross-sectional features. Section 5 (Results) presents the out-of-sample results, the cross-sectional z-score and cluster analysis that documents the regime-momentum mechanism, robustness and risk-aversion analyses, and international replication. Section 6 (Summary, Conclusions and Recommendations) summarises the contributions, discusses limitations, and suggests directions for future research.

2 Literature Review

This review covers four strands of literature: the cross-sectional momentum anomaly and its horizon structure, momentum crashes and exposure-based risk management, regime models and their use in regime-conditional asset pricing, and the application of machine learning to cross-sectional return prediction. Each provides a piece of the framework this thesis assembles.

2.1 The Momentum Anomaly and Its Horizon Structure

Jegadeesh and Titman (1993) established cross-sectional momentum as a pervasive feature of equity markets. Portfolios that buy past winners and sell past losers earn significant returns over 3-to-12-month holding periods. They examine sixteen formation-and-holding-period combinations from three to twelve months, with the twelve-month formation period emerging as the most profitable. The strategy has since been replicated across international markets, asset classes, and sample periods (Jegadeesh and Titman, 2001; Asness et al., 2013).

The economic mechanism behind momentum remains debated. Behavioural explanations include gradual information diffusion (Hong and Stein, 1999) and the joint conservatism-representativeness mechanism of Barberis et al. (1998), while risk-based explanations link momentum to time-varying systematic risk. This thesis does not take a position on the underlying mechanism but examines the empirical regularity that momentum's cross-sectional structure varies with market conditions.

A defining feature of momentum is its sensitivity to the lookback horizon. Short lookbacks of one to three months capture short-term reversal effects (Jegadeesh, 1990; Lehmann, 1990), intermediate horizons of seven to twelve months prior to formation capture continuation (Novy-Marx, 2012), and lookbacks beyond twelve months reflect long-term mean reversion (De Bondt and Thaler, 1985). Lee and Swaminathan (2000) document a momentum lifecycle in which the same signal evolves from continuation to reversal as information ages. Together these findings suggest that momentum is a family of horizon-specific patterns whose relative strength varies with market conditions, not a single signal. The framework here tests this directly by exposing all twelve monthly horizons jointly to the cross-sectional model rather than committing to a single lookback ex ante.

2.2 Momentum Crashes and Exposure-Based Solutions

Despite its long-run profitability, momentum is prone to severe crashes. Daniel and Moskowitz (2016) document that momentum portfolios suffered catastrophic losses during the 1932 and 2009 market rebounds, and show that these crashes are predictable: when the market has recently declined, past losers tend to be high-beta stocks that rally disproportionately during recoveries, while past winners are low-beta defensive stocks that lag. In economic terms, momentum is a bet that the recent regime persists. It earns the continuation premium when the regime holds, and crashes at regime transitions, when the cross-sectional ordering inherited from the prior regime becomes the wrong bet for the new one. Barroso and Santa-Clara (2015) propose a simple risk-management response, scaling momentum exposure inversely with the strategy's recent realised volatility, a constant-volatility overlay that substantially raises the Sharpe ratio. Bianchi et al. (2022) pursue a related approach by constructing a crash indicator from the joint time variation of conditional volatility and skewness, again operating on the strategy's exposure rather than its composition. This thesis explores a complementary composition-side route. Rather than scaling exposure during stress, the model optimizes on which stocks populate each leg conditional on the regime, with the consequence that the leg-beta inversion underlying the crash mechanism does not occur (Appendix H.5).

The liquidity literature provides a theoretical basis for why momentum crashes reverse. Margin spirals (Brunnermeier and Pedersen, 2009) and institutional flow pressure (Vayanos and Woolley, 2013) create temporary price dislocations during stress that revert when selling pressure subsides, the mechanism central to the regime-momentum interpretation developed below.

2.3 Regime Models and Regime-Conditional Asset Pricing

Hidden Markov Models have a long history in econometrics: the underlying probabilistic machinery dates to Baum et al. (1970), and Hamilton (1989) introduced the regime-switching tradition with a business-cycle application to U.S. GNP growth. Later extended to Bayesian estimation via Gibbs sampling by Albert and Chib (1993), with the multivariate-emission generalisation now standard (Frühwirth-Schnatter, 2006). A common finding in the regime-switching literature for equity markets is that returns are well described by a two-state model with a high-mean, low-volatility state and a low-mean, high-volatility state, where the bear state is less persistent but more volatile (Ang and Bekaert, 2002; Guidolin and Timmermann, 2007). Hamilton's original application fitted the regime model to GNP growth rather than returns themselves, a design choice followed here by using macrofinancial stress indicators as observable inputs.

A separate strand asks whether asset-pricing anomalies are themselves regime-conditional. Cooper et al. (2004) establish that momentum profits depend on the prior market state, with strong continuation following up-markets and weaker or reversed performance following down-markets. More recent work integrates regime estimation into portfolio construction: Shu and Mulvey (2024) use a sparse jump model to identify factor-specific bull and bear regimes and feed the resulting classifications into a Black-Litterman framework for dynamic factor allocation.

The most directly related work is Goulding et al. (2023), who classify each month into one of four market cycles (Bull, Correction, Bear, Rebound) from the trailing one-month and rolling twelve-month market returns. In each cycle a different blending parameter a determines the weight on one-month versus twelve-month momentum: the composite signal is $a \cdot mom_1 + (1 - a) \cdot mom_{12}$, where $a = 0$ corresponds to pure twelve-month momentum and $a = 1$ to pure one-month momentum. Their dynamic strategy grid-searches the optimal blending parameters a_{Co} and a_{Re} for the two turning-point states (Correction and Rebound) on the training sample. In Bull and Bear states, the slow and fast signals agree, so any value of a produces the same position. The framework here generalises that approach along three axes: from a deterministic four-state classification based on trailing returns to a continuous panic probability from a Bayesian HMM on real-time stress indicators. From two horizons to the full twelve-horizon term structure. And from a parametric linear blend to a learned nonlinear combination. Each of these moves is necessary, individually, to pose a question GHM cannot: whether the cross-sectional structure of momentum reorganises across regimes. Because GHM's strategies are a time-series optimization of a single market index rather than a cross-sectional long-short of individual stocks, a direct performance comparison is not appropriate; Section 5 instead uses the GHM market-cycle variable as an alternative regime input within the XGBoost framework, providing a fair test of whether trailing-return-based regime definitions capture the same information as the HMM's stress indicators.

Furthermore, the unifying feature of this strand is that the regime variable enters portfolio decisions through a hand-coded rule, threshold, or parametric weight, never as a continuous learned feature. This thesis treats π_t^{filter} instead as an input passed to a learner that decides how to use it.

2.4 Machine Learning in Cross-Sectional Asset Pricing

The use of machine learning methods for cross-sectional stock selection has grown rapidly. Gu et al. (2020) compare a wide range of methods on a kitchen-sink characteristic set including multiple momentum horizons, finding that tree-based models and neural networks substantially outperform linear models by capturing nonlinear interactions among firm characteristics. Gradient-boosted trees (Chen and Guestrin, 2016) have proven particularly effective for tabular financial data because they handle missing values natively, capture nonlinear relationships and interactions without explicit specification, and are robust to irrelevant features. A central challenge in applying these methods to asset pricing is interpretability, addressed here through SHAP values (Lundberg and Lee, 2017; Lundberg et al., 2020), which decompose each prediction into feature-level contributions and admit an exact polynomial-time algorithm for tree ensembles.

The closest methodological precedent is Beckmeyer and Wiedemann (2025), who use machine learning to learn data-driven weights across past daily returns for a single momentum signal, constructing a dynamically reweighted strategy that outperforms the conventional twelve-minus-one specification. Their finding establishes that data-driven weighting of past returns outperforms fixed schemes. The framework here differs in two respects. First, the weighting is learned not over the daily history of one signal but over a vector of twelve monthly horizons. Second, and more importantly, the weighting is learned *conditional on a regime context variable*, isolating the regime-conditioning role as the primary contribution. The unconditional weighting in Beckmeyer and Wiedemann (2025) averages across calm and panic regimes, while this thesis tests whether disaggregating that average reveals a structural reorganisation of the term structure between states.

Despite these advances, no published study examines how the full term structure of cross-sectional momentum predictability reorganises across regimes, or whether capturing this reorganisation requires nonlinear modelling.

Position of This Thesis

Each strand contributes one ingredient that the others cannot supply on their own. The momentum literature provides the object of study, a family of horizon-specific signals whose relative strength is known to vary with conditions but has been treated unconditionally in standard practice. The crash literature documents the cost of this fragility, shows that exposure-scaling overlays can partially manage it, and motivates examining the stock-selection level alongside the exposure level when characterising momentum's regime-dependence. The regime-conditional asset-pricing literature provides the conditioning variable, although it has consistently been used as a switch or parametric weight rather than as a continuous learned feature. The machine-learning literature provides the function class capable of representing high-dimensional interactions, although its existing applications have addressed kitchen-sink characteristic sets or unconditional momentum weighting rather than regime-conditional momentum structure.

Section 3 develops the two-stage framework that assembles these four ingredients.

3 Methodology

This chapter develops the two-stage framework used to test the research question posed in Section 1. Section 3.1 specifies a Bayesian two-state Hidden Markov Model on four macrofinancial stress indicators, derives the Gibbs sampling algorithm used for posterior inference, and identifies the resulting filtered probability $\pi_t^{\text{filter}} = P(s_t = 1 \mid \mathbf{z}_{1:t})$ as the regime signal, a probabilistic projection of the four-dimensional stress trajectory onto a one-dimensional regime axis. Section 3.2 pairs the regime signal with twelve momentum horizons in three cross-sectional stock-selection methods of increasing complexity (a deterministic formula DET, a logistic regression LR, and an XGBoost regressor XGB), whose comparison provides the key diagnostic for the regime-momentum interaction. If the regime signal functions as a context variable rather than a direct return predictor, a linear model should be unable to exploit it while a tree-based model should find it important.

3.1 Stage 1: Regime Classification via Hidden Markov Model

3.1.1 Model Specification

This thesis employs a two-state ($K = 2$) Bayesian Hidden Markov Model to classify the market into latent regimes. The choice of $K = 2$ is validated empirically. Appendix G.4 shows that increasing to $K = 3, 4$, or 5 states does not improve downstream portfolio performance and introduces estimation instability. The hidden state $s_t \in \{0, 1\}$ represents the unobserved regime at month t, where $s_t = 0$ denotes the *calm* regime and $s_t = 1$ denotes the *panic* regime. The labels *calm* and *panic* reflect the economic nature of the chosen stress features (drawdown, return dispersion, relative market participation, and the credit spread), which are most directly diagnostic of market stress rather than of trend direction, business-cycle phase, or volatility alone. A HMM fitted to a different feature set would identify different regimes. Crucially, as emphasized by Hamilton (1989), market regimes are not directly observable. An investor cannot know with certainty whether the market is currently in a calm or panic state, and classifications that look obvious in retrospect are genuinely difficult to identify in real time. The HMM formalises this uncertainty by treating regimes as hidden states that must be inferred probabilistically from observable market data (see Figure 1).

Observation equation

Conditional on the regime, the vector of standardized features $\mathbf{z}_t \in \mathbb{R}^D$ (where $D = 4$) is drawn from a regime-specific multivariate Gaussian. The shorthand $\mathbf{z}_{1:t} = (\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_t)$

denotes the full history of observations up to and including month t:

$$(\mathbf{z}_t \mid s_t = 0) \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), \qquad (\mathbf{z}_t \mid s_t = 1) \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) \tag{1}$$

Each regime is characterized by its own mean vector $\boldsymbol{\mu}_k \in \mathbb{R}^D$ and covariance matrix $\boldsymbol{\Sigma}_k \in \mathbb{R}^{D \times D}$, where $k \in \{0 \text{ (calm)}, 1 \text{ (panic)}\}$. The multivariate specification allows the model to capture not only level differences in individual features across regimes, but also changes in the correlation structure. For instance, stress features may become more correlated during panic periods. The Gaussian assumption is standard in the HMM literature. Appendix H.8 verifies its adequacy by re-estimating the model with heavier-tailed multivariate Student-t emissions across a grid of degrees of freedom, the natural alternative if monthly stress features carry tail risk that a Gaussian under-weights. The two specifications produce essentially the same regime signal: the binary calm/panic classification matches the Gaussian model in at least 97.8% of months, the month-by-month filtered panic probabilities correlate above 0.98, and the Gaussian model achieves a marginally lower BIC. Heavier tails therefore neither change the inferred regime structure nor improve fit, confirming that the Gaussian specification is adequate for this application.

Transition equation

The evolution of the hidden state follows a first-order Markov chain with a 2×2 transition matrix:

$$\mathbf{P} = \begin{pmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{pmatrix} \tag{2}$$

where $P_{ij} = P(s_t = j \mid s_{t-1} = i)$ and each row sums to one. The diagonal elements P_{00} (calm persistence) and P_{11} (panic persistence) govern how likely each regime is to continue into the next month. The expected duration of the calm regime is $1/(1 - P_{00})$ months, and likewise $1/(1 - P_{11})$ for the panic regime.

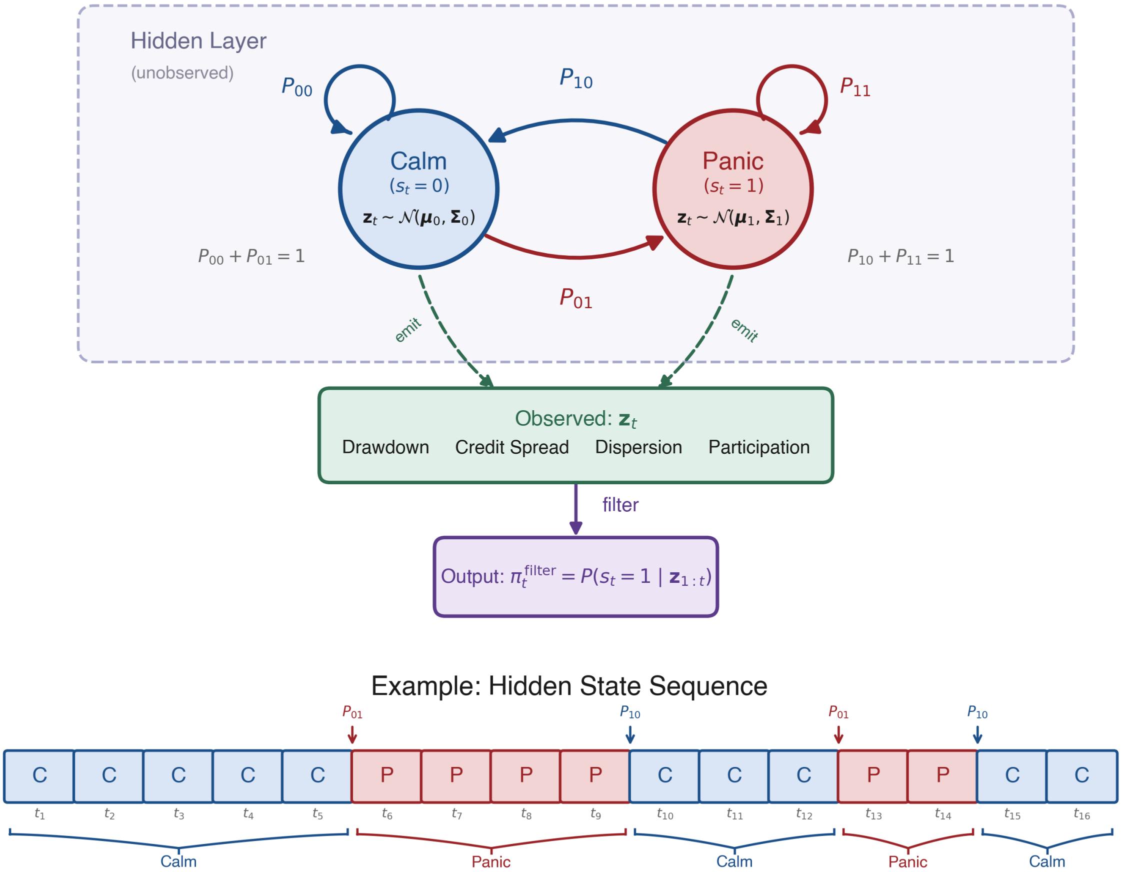


Figure 1: Architecture of the two-state Hidden Markov Model. Hidden states (Calm/Panic) evolve via transition probabilities P_{ij} and emit the four-dimensional observation vector $\mathbf{z}_t$. The forward filter inverts this process to produce $\pi_t^{\text{filter}} = P(s_t = 1 \mid \mathbf{z}_{1:t})$. The lower panel shows an example state sequence illustrating regime persistence and abrupt transitions.

3.1.2 Bayesian Estimation via Gibbs Sampling

The standard frequentist alternative is the Baum–Welch algorithm (Baum et al., 1970), a special case of Expectation-Maximisation (Dempster et al., 1977), which finds maximum likelihood point estimates.

Maximum likelihood estimation of regime-switching HMMs has several known shortcomings for the present application. The likelihood surface is multimodal, with at minimum two equivalent global optima arising from the invariance of the likelihood under state relabelling. Additionally, expectation maximization can converge to degenerate solutions in which one regime collapses around a single observation (Frühwirth-Schnatter, 2006). Both problems are exacerbated for rare regimes, the panic state in this sample contains only 91 monthly observations in training, where the asymptotic standard errors that frequentist inference relies on are unreliable, and the standard regularity conditions can fail because the persistence parameters P_{00} and P_{11} are unidentified when the two regimes' means coincide and the information matrix becomes singular at that point (Hamilton, 1989, footnote 12). The Bayesian approach addresses these issues directly. Posterior sampling integrates over parameter uncertainty rather than relying on a point estimate. Conjugate priors regularise rare-regime estimates with explicit, defensible weights (Section 3.1.3). And convergence diagnostics across multiple chains (Section 3.1.5) diagnose multimodality that would otherwise go unnoticed under a single EM run.

The Bayesian approach instead draws from the joint posterior:

$$p\big(\{s_t\}_{t=1}^{T},\ \{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=0}^{1},\ \mathbf{P} \,\big|\, \mathbf{z}_{1:T}\big)$$

The states and parameters are coupled, in the sense that each block has a tractable distribution conditional on the other but neither has a tractable marginal. Gibbs sampling (Albert and Chib, 1993; Hoff, 2009), a Markov chain Monte Carlo (MCMC) method, exploits exactly that structure by alternating between three conditional updates: (i) the full hidden state path $\{s_t\}_{t=1}^{T}$ (sampled with the forward-filtering backward-sampling algorithm given parameters), (ii) the emission parameters $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ for each regime (given the state path), and (iii) the transition matrix $\mathbf{P}$ (given the state path). Conjugate priors ensure that each block can be sampled exactly from its conditional posterior.

3.1.3 Prior Specification

The priors encode two economic beliefs. First, before seeing the data, neither regime should be forced to have extreme values in any feature; the model should learn the characteristics of calm and panic states from the data. Second, regimes should be persistent rather than switching every few months.

Emission parameters

For each regime k, the mean vector and covariance matrix are given a conjugate Normal-Inverse-Wishart (NIW) prior:[3]

$$\boldsymbol{\Sigma}_k \sim \mathcal{IW}(\nu_0, \boldsymbol{\Psi}_0), \qquad \boldsymbol{\mu}_k \mid \boldsymbol{\Sigma}_k \sim \mathcal{N}\left(\mathbf{m}_0, \frac{\boldsymbol{\Sigma}_k}{\kappa_0}\right) \tag{3}$$

The hyperparameters are chosen as

$$\mathbf{m}_0 = \mathbf{0} \qquad \kappa_0 = 0.01 \qquad \nu_0 = D + 2 \qquad \boldsymbol{\Psi}_0 = \mathbf{I}_D(\nu_0 - D - 1) \tag{4}$$

[3]The Inverse-Wishart is a probability distribution over positive-definite matrices, making it the natural prior for a covariance matrix. Every sample is guaranteed to be symmetric and positive-definite. The Normal-Inverse-Wishart combines this with a conditional Normal prior on the mean. The prior is called *conjugate* because when the data are Gaussian, the posterior takes the same NIW form with updated hyperparameters, making each Gibbs update an exact draw rather than an approximation.

where $D = 4$ is the number of features. After observing data, these prior hyperparameters update to posterior counterparts κ_n, $\mathbf{m}_n$, ν_n, and $\mathbf{\Psi}_n$ (Equation 9). Each hyperparameter controls a different aspect of the prior belief:

- $\mathbf{m}_0$ is the prior guess for each regime's mean vector. $\mathbf{m}_0 = \mathbf{0}$ is the natural prior given that the features are standardised. Without data, neither regime is presumed to centre away from zero.

- κ_0 controls how strongly the prior mean $\mathbf{m}_0$ is trusted relative to the data. The very small value $\kappa_0 = 0.01$ makes this prior nearly uninformative, so the posterior mean is driven overwhelmingly by the observed data.

- ν_0 is the degrees of freedom of the Inverse-Wishart prior, interpretable as the number of pseudo-observations the prior is worth. The value $\nu_0 = D + 2 = 6$ is the smallest integer that yields a proper prior with finite mean (any smaller value gives an improper prior). A larger choice (e.g., $\nu_0 = 9$) would impose stronger confidence that the covariance is close to $\mathbf{I}_D$. With roughly 150 calm and 91 panic months in training, the posterior degrees of freedom ($\nu_n = \nu_0 + n_k$) far exceed the prior, so the data dominate the covariance estimates.

- $\mathbf{\Psi}_0$ is the scale matrix of the Inverse-Wishart prior. It is calibrated so that the prior expected covariance equals the identity matrix, $\mathbb{E}[\mathbf{\Sigma}_k] = \mathbf{\Psi}_0/(\nu_0 - D - 1) = \mathbf{I}_D$, implying unit marginal variance and no cross-feature correlation before observing the data.

Transition matrix

Each row of the transition matrix is assigned an independent Dirichlet prior:

$$\mathbf{P}_{0,\cdot} \sim \mathrm{Dir}(9, 1) \qquad \mathbf{P}_{1,\cdot} \sim \mathrm{Dir}(1, 9) \tag{5}$$

Equivalently, in matrix form,

$$\boldsymbol{\alpha}_{\mathrm{dir}} = \begin{pmatrix} 9 & 1 \\ 1 & 9 \end{pmatrix}$$

This prior favors persistence in both regimes. Without data, the prior mean transition matrix is

$$\mathbb{E}[\mathbf{P}] = \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix} \tag{6}$$

In other words, the prior expects each regime to continue with probability 0.9, corresponding to an expected duration of roughly 10 months. This prior is informative enough to discourage pathological solutions but weak enough to be overridden by the data.

3.1.4 Gibbs Sampling Algorithm

Initialization

The Gibbs sampler requires starting values for the emission parameters and the transition matrix. A crude drawdown-based partition is used:[4] months with below-median drawdown DD_t^z (i.e., deeper drawdowns) are initially assigned to the panic state ($s_t = 1$), and the remaining months to the calm state ($s_t = 0$). The initial emission parameters are then computed from each group:

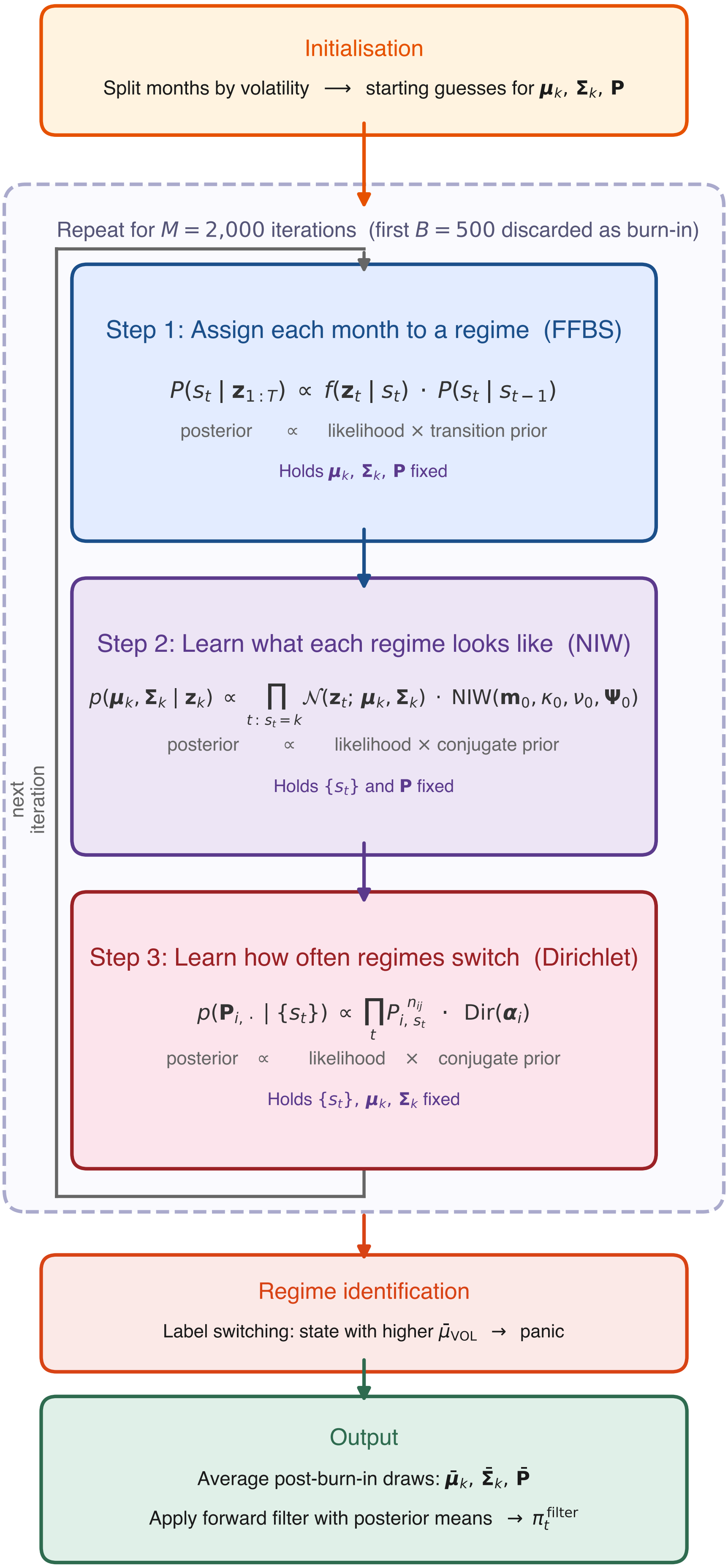
$$\boldsymbol{\mu}_k^{(0)} = \frac{1}{n_k} \sum_{t:\, s_t = k} \mathbf{z}_t \qquad \boldsymbol{\Sigma}_k^{(0)} = \mathrm{Cov}(\{\mathbf{z}_t : s_t = k\})$$

where n_k is the number of months assigned to regime k. The initial transition matrix is set to the prior mean, $\mathbf{P}^{(0)} = \mathbb{E}[\mathbf{P}]$ (Equation 6). These initial parameters give the first forward pass (described below) emission densities that can distinguish the two states.

Iterative updates

Each iteration cycles through the three conditional updates described in Section 3.1.2. Figure 2 summarizes the cycle graphically.

[4] The choice of initialization has negligible impact on the posterior. The sampler is run for 2,000 iterations with 500 discarded as burn-in, and convergence diagnostics (Section 3.1.5) confirm that the chain mixes well regardless of the starting partition.


Initialisation
Split months by volatility ⟶ starting guesses for $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$, $\mathbf{P}$
Repeat for $M = 2{,}000$ iterations (first $B = 500$ discarded as burn-in)
next iteration
Step 1: Assign each month to a regime (FFBS)
$P(s_t \mid \mathbf{z}_{1:T}) \propto f(\mathbf{z}_t \mid s_t) \cdot P(s_t \mid s_{t-1})$
posterior ∝ likelihood × transition prior
Holds $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$, $\mathbf{P}$ fixed
Step 2: Learn what each regime looks like (NIW)
$p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k \mid \mathbf{z}_k) \propto \prod_{t:\, s_t = k} \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \cdot \mathrm{NIW}(\mathbf{m}_0, \kappa_0, \nu_0, \boldsymbol{\Psi}_0)$
posterior ∝ likelihood × conjugate prior
Holds $\{s_t\}$ and $\mathbf{P}$ fixed
Step 3: Learn how often regimes switch (Dirichlet)
$p(\mathbf{P}_{i,\cdot} \mid \{s_t\}) \propto \prod_t P_{i,\, s_t}^{n_{ij}} \cdot \mathrm{Dir}(\boldsymbol{\alpha}_i)$
posterior ∝ likelihood × conjugate prior
Holds $\{s_t\}$, $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$ fixed
Regime identification
Label switching: state with higher $\bar{\mu}_{\mathrm{VOL}}$ → panic
Output
Average post-burn-in draws: $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, $\bar{\mathbf{P}}$
Apply forward filter with posterior means → π_t^{filter}

Step 1: Sampling the hidden state path via FFBS

The full latent regime sequence $(s_1, \ldots, s_T)$ is sampled using Forward-Filtering Backward-Sampling (FFBS; see Frühwirth-Schnatter, 2006, ch. 11), holding the current emission parameters and transition matrix fixed.

The forward pass computes the filtered probability $\alpha_t(k) = P(s_t = k \mid \mathbf{z}_{1:t})$, the probability of being in state k at time t given all observations so far. This quantity is built in two sub-steps. First, a *prediction step* asks: before seeing this month's data, how likely is each state? Since the previous state is unknown, the algorithm sums over both possibilities, weighting each by its filtered probability from the previous month and the transition probability:

$$\alpha_{t|t-1}(k) = \alpha_{t-1}(0)\, P_{0k} + \alpha_{t-1}(1)\, P_{1k}$$

Second, an *update step* revises this prediction once the current observation $\mathbf{z}_t$ arrives, using Bayes' rule. If the observed features look more like a panic month than a calm month, the probability of panic increases. Combining the two steps yields the recursion

$$\alpha_t(k) = \frac{f(\mathbf{z}_t \mid s_t = k)\, \alpha_{t|t-1}(k)}{f(\mathbf{z}_t \mid s_t = 0)\, \alpha_{t|t-1}(0) + f(\mathbf{z}_t \mid s_t = 1)\, \alpha_{t|t-1}(1)} \tag{7}$$

where $f(\mathbf{z}_t \mid s_t = k) = \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is the Gaussian emission density.[5] The recursion is initialised with a uniform initial state distribution $P(s_1 = k) = 1/2$, reflecting no prior preference for either regime. For the first observation this gives

$$\alpha_1(k) \propto f(\mathbf{z}_1 \mid s_1 = k)$$

The forward pass tells us how likely each regime is at every month, but if we simply flipped a weighted coin independently at each t we could produce implausible paths (for example, rapid calm-panic-calm-panic switching even though the transition matrix says regimes persist for roughly ten months). The backward pass prevents this by sampling states in reverse. Once a future state has been drawn, the current state is sampled conditional on it, so the entire path respects the Markov transition dynamics.

The backward pass proceeds in two steps:

Terminal state

At $t = T$, the filtered probability already conditions on all observations, so the terminal state is drawn directly:

$$s_T \sim \mathrm{Cat}(\alpha_T(0),\, \alpha_T(1))$$

[5] In practice, these densities can be extremely small (or large) for four-dimensional observations, leading to numerical underflow. The implementation avoids this by working in log-space. Log-likelihoods $\log f(\mathbf{z}_t \mid s_t = k)$ are computed directly, and the log-sum-exp identity $\log(e^a + e^b) = \max(a, b) + \log(1 + e^{-|a-b|})$ is used whenever probabilities must be normalised.

that is, the algorithm draws $s_T = 0$ (calm) with probability $\alpha_T(0)$ and $s_T = 1$ (panic) with probability $\alpha_T(1)$.[6]

Backward recursion

For each earlier time step $t = T-1, T-2, \ldots, 1$, the algorithm chooses a state consistent with two pieces of information: (i) the filtered belief $\alpha_t(k)$ from the forward pass, and (ii) the already-sampled future state s_{t+1}, which constrains the current state via the transition probability $P_{k,s_{t+1}}$. Combining these two terms and normalising gives:

$$P(s_t = k \mid s_{t+1}, \mathbf{z}_{1:t}) = \frac{\alpha_t(k)\, P_{k,s_{t+1}}}{\alpha_t(0)\, P_{0,s_{t+1}} + \alpha_t(1)\, P_{1,s_{t+1}}} \tag{8}$$

For example, if the forward filter assigns high probability to calm at time t ($\alpha_t(0)$ is large) and the already-sampled s_{t+1} is also calm (so $P_{0,0}$ is large), then the numerator for $k = 0$ dominates and the algorithm will very likely sample $s_t = 0$.

The output of the full FFBS procedure is one draw from the posterior over state paths. Across Gibbs iterations, different paths are sampled, exploring the full posterior uncertainty.

Importantly, the backward pass is used only during training: the sampled state path it produces feeds the conjugate Gibbs updates of the emission and transition parameters. The trading signal π_t^{filter} relies solely on the causal forward filter.

Step 2: Updating the emission parameters

Given the sampled state path, each regime k has n_k observations assigned to it. From these, two summary statistics are computed:

$$\bar{\mathbf{x}}_k = \frac{1}{n_k} \sum_{t:\, s_t = k} \mathbf{z}_t \qquad \mathbf{S}_k = \sum_{t:\, s_t = k} (\mathbf{z}_t - \bar{\mathbf{x}}_k)(\mathbf{z}_t - \bar{\mathbf{x}}_k)^\top$$

The posterior hyperparameters are:

$$\kappa_n = \kappa_0 + n_k \qquad \mathbf{m}_n = \frac{\kappa_0 \mathbf{m}_0 + n_k \bar{\mathbf{x}}_k}{\kappa_n} \qquad \nu_n = \nu_0 + n_k$$

$$\boldsymbol{\Psi}_n = \boldsymbol{\Psi}_0 + \mathbf{S}_k + \frac{\kappa_0 n_k}{\kappa_n} (\bar{\mathbf{x}}_k - \mathbf{m}_0)(\bar{\mathbf{x}}_k - \mathbf{m}_0)^\top \tag{9}$$

Since $\kappa_0 = 0.01 \ll n_k$, the posterior is driven almost entirely by the data.

New regime parameters are then drawn from the updated posterior: $\boldsymbol{\Sigma}_k \sim \mathcal{IW}(\nu_n, \boldsymbol{\Psi}_n)$ and $\boldsymbol{\mu}_k \mid \boldsymbol{\Sigma}_k \sim \mathcal{N}(\mathbf{m}_n, \boldsymbol{\Sigma}_k/\kappa_n)$.

[6] Cat denotes the categorical distribution, the standard notation for a single draw from a discrete probability vector. With two states it reduces to a weighted coin flip.

Step 3: Updating the transition matrix

The sampled path is scanned to count transitions: $n_{ij} = \sum_{t=2}^{T} \mathbb{1}(s_{t-1} = i,\, s_t = j)$ records how many times the path moved from state i to state j. For example, if the sampled path contains 140 calm-to-calm transitions and 10 calm-to-panic transitions, then $n_{00} = 140$ and $n_{01} = 10$. These observed counts are added to the Dirichlet prior pseudo-counts to obtain the posterior:

$$\mathbf{P}_{i,\cdot} \mid \{s_t\} \sim \mathrm{Dir}\big(\alpha_{i0} + n_{i0},\ \alpha_{i1} + n_{i1}\big) \tag{10}$$

For row 0 (calm), the prior pseudo-counts are $(\alpha_{00}, \alpha_{01}) = (9, 1)$ from Equation (5), so the posterior becomes $\mathrm{Dir}(9 + 140,\ 1 + 10) = \mathrm{Dir}(149,\ 11)$. The expected persistence probability under this posterior is $149/160 = 0.93$, reflecting that the data reinforced the prior belief in regime persistence. A new transition matrix row is then drawn from this Dirichlet distribution, and the same procedure is applied independently to row 1 (panic).

With all three blocks updated, the sampler returns to Step 1. After sufficient iterations the draws converge to samples from the joint posterior.

3.1.5 Implementation and Convergence

The Gibbs sampler is estimated on the training sample only. To reduce sensitivity to MCMC initialization, the full sampler is run independently with $N_s = 200$ different random seeds. Each chain is run for $M = 2{,}000$ iterations with the initialization described above. The first $B = 500$ iterations are discarded as burn-in, leaving 1,500 posterior draws per seed.

For each seed, posterior mean parameters $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, and $\bar{\mathbf{P}}$ are computed by averaging over the post-burn-in draws. A causal forward filter is then applied to the full sample using these parameters to obtain a seed-specific filtered probability $\pi_{t,s}^{\text{filter}}$. The final trading signal is the average across seeds:

$$\pi_t^{\text{filter}} = \frac{1}{N_s} \sum_{s=1}^{N_s} \pi_{t,s}^{\text{filter}} \tag{11}$$

This averaging across MCMC chains is a methodological requirement, not a performance optimisation. With a single Gibbs chain the out-of-sample Sharpe has mean 1.04 and interquartile range 0.10 across random HMM seed draws, while averaging 50 chains raises the mean to 1.09 and collapses the IQR to 0.02 (Appendix J.2). The 1.11 figure reported in Section 5 therefore reflects the production 200-seed configuration and should be interpreted as an ensemble-averaged quantity across chain initialisations rather than a single-chain point estimate.

Convergence Diagnostics

Convergence is assessed in two ways. First, trace plots of the regime means and persistence probabilities are visually inspected to verify that the chain mixes around a stable posterior region after burn-in. Second, effective sample sizes (ESS) are computed for key parameters. Because successive MCMC draws are autocorrelated, n draws from the chain contain less information than n independent samples. The ESS quantifies the number of truly independent draws the chain is equivalent to. For a post-burn-in chain of length n,

$$ESS = \frac{n}{1 + 2\sum_{\ell=1}^{L^*} \hat{\rho}(\ell)} \tag{12}$$

where $\hat{\rho}(\ell)$ is the estimated autocorrelation at lag ℓ,[7] and L^* is the first lag at which the autocorrelation becomes negative. High ESS indicates that the chain provides many effectively independent posterior draws.[8]

While ESS measures efficiency within a single chain, it cannot detect whether different chains have converged to different modes. The Gelman–Rubin $\hat{R}$ statistic (Gelman and Rubin, 1992; Gelman et al., 2013) addresses this by comparing within-chain and between-chain variance across M independent chains, each of length n. Let s_m^2 denote the variance of chain m and $\bar{\theta}_m$ its mean. The within-chain variance is:

$$W = \frac{1}{M} \sum_{m=1}^{M} s_m^2 \tag{13}$$

and the between-chain variance is:

$$B = \frac{n}{M-1} \sum_{m=1}^{M} (\bar{\theta}_m - \bar{\theta})^2 \tag{14}$$

where $\bar{\theta} = \frac{1}{M}\sum_m \bar{\theta}_m$. If all chains have converged to the same posterior, B should be small relative to W. The diagnostic is:

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}}, \qquad \hat{V} = \frac{n-1}{n} W + \frac{B}{n} \tag{15}$$

Values of $\hat{R}$ near 1.0 indicate convergence. The conventional threshold is $\hat{R} < 1.1$, with a stricter modern bound at $\hat{R} < 1.01$. To evaluate convergence, five of the 200 chains are selected at random. For the selected feature set (DD+DISP+REL_N+CS), all parameters have $\hat{R} < 1.01$ (Appendix J.1).

[7] $\hat{\rho}(\ell) = \frac{1}{(n-\ell)\,\hat{\sigma}^2} \sum_{i=1}^{n-\ell} (\theta_i - \bar{\theta})(\theta_{i+\ell} - \bar{\theta})$, where θ_i is the i-th post-burn-in draw, $\bar{\theta}$ its sample mean, and $\hat{\sigma}^2$ its sample variance.

[8] An ESS of at least 100 per parameter is a common minimum for reliable posterior inference (Gelman et al., 2013).

Trace plots of the regime means and persistence probabilities (Figure 13, Appendix J) provide visual confirmation that all chains converge to stable posterior values after the burn-in period.

3.1.6 Regime Identification and Trading Signal

Label Switching and Regime Naming

In a two-state HMM, the labels 0 and 1 are intrinsically arbitrary. Relabelling the states leaves the likelihood unchanged. The HMM therefore identifies two clusters in the four-dimensional feature space without knowing which one should be called panic. To break the symmetry, the feature behaviour during two externally-defined historical crisis windows (the dot-com bust of 2000–2002 and the Global Financial Crisis of 2007–2009) is used as a reference. In those windows DISP and CS sit at higher levels and DD and REL_N sit at lower levels than in the rest of the training sample, yielding sign assignments DISP $= +1$, CS $= +1$, DD $= -1$, REL_N $= -1$ (Appendix I). The cluster whose posterior mean vector best aligns with that pattern is designated as "panic" and the other as "calm". If the Gibbs sampler assigns the panic characteristics to state 0 rather than state 1, the labels are swapped so that π_t^{filter} always represents the probability of the panic state. The crisis windows enter only as this one-shot labelling step. The HMM never sees them, and they do not define what the panic state is, only which of the two emergent clusters carries that name.

Two caveats follow from this naming convention. The HMM identifies two regimes by clustering on the emission features. The label "panic" is then assigned to the cluster whose posterior mean aligns with crisis directions. The label therefore reflects a statistical resemblance to historical crisis periods rather than a directly observed market state. Relatedly, the economic interpretations used later in the thesis (liquidity pressure, constrained arbitrage) are readings of what this cluster *likely* corresponds to given the features it loads on, not direct identification claims about the underlying mechanism.

Filtered Regime Probability

Using the posterior mean parameters $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, and $\bar{\mathbf{P}}$ from Section 3.1.5, the forward recursion (Equation 7) is applied to the full sample to obtain:

$$\pi_t^{\text{filter}} = P(s_t = \text{panic} \mid \mathbf{z}_{1:t};\, \bar{\boldsymbol{\mu}}, \bar{\boldsymbol{\Sigma}}, \bar{\mathbf{P}}) \tag{16}$$

A value near 0 indicates high confidence in calm, near 1 indicates panic, and intermediate values reflect genuine ambiguity. This filtered probability serves as the regime signal passed to the portfolio construction methods in Section 3.2.1. Figure 4 and Table 2 in Section 4 visualise the resulting signal and confirm that the two regimes are meaningfully distinct.

3.2 Stage 2: Regime-Conditioned Portfolio Construction

The filtered regime probability π_t^{filter} from Stage 1 provides a monthly measure of market stress, but on its own it does not select stocks. This stage translates the regime signal into cross-sectional portfolios. Three cross-sectional stock selection methods of increasing complexity are developed: a deterministic regime-adaptive momentum formula (DET), a logistic regression classifier (LR), and an XGBoost gradient-boosted tree regressor (XGB). Each method combines π_t^{filter} with momentum signals at multiple horizons to produce a monthly score for every stock, which is then used to rank stocks into a long-short portfolio: the top decile (using NYSE breakpoints) forms the long leg, the bottom decile forms the short leg, and both legs are value-weighted. The stock-level inputs (momentum horizons and the regime signal) are described in Section 4.

A key distinction from standard momentum strategies is worth emphasising. In the traditional approach (Jegadeesh and Titman, 1993), stocks are ranked by their raw trailing return at a fixed lookback, so the long leg always holds the highest-momentum stocks and the short leg the lowest. Here, stocks are ranked by their predicted forward return, a learned function of the regime signal and the full momentum term structure, so the long leg need not contain high-momentum stocks at all. In panic, it can hold stocks below the cross-sectional average at every horizon (Section 5.2.2). The portfolio construction remains identical (top and bottom deciles), but the criterion for entering each decile is regime-dependent rather than fixed. The deterministic formula (DET) serves as a transparent baseline. The two machine learning methods go further by jointly learning the regime-momentum interaction.

We evaluate every method in long-short construction rather than long-only, for three reasons. First, the long-short portfolio is dollar-neutral by construction ($1 long the top decile, $1 short the bottom decile, both value-weighted), so the realised return reflects whether top-decile stocks outperform bottom-decile stocks rather than market direction. Second, the mechanism is two-sided. In panic the model predicts reversal, identifying stocks expected to fall as well as those expected to rise, and the short leg directly tests whether those downward predictions are informative. Third, long-short is the standard construction in the momentum literature (Jegadeesh and Titman, 1993; Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015), which makes our Sharpe ratios and factor alphas directly comparable to existing results. The value-weighted market (long-only, $\beta = 1$) is reported alongside the long-short rows as a passive equity benchmark, but is not directly Sharpe-comparable with the long-short strategies, since it harvests the equity premium whereas the long-short rows earn their return from cross-sectional ranking. The proper market-relative test that controls for exposure is the six-factor alpha (Appendix H.3).

3.2.1 Cross-Sectional Stock Selection Methods

DET: Deterministic Regime-Adaptive Formula

The simplest approach exploits the economic intuition that the optimal momentum lookback horizon varies with market conditions.[9] The lookback horizon and stock score are:

$$\ell_t = \text{round}\big(12 - 11 \cdot \pi_t^{\text{filter}}\big) \tag{17}$$

$$\text{score}_t^s = \prod_{j=1}^{\ell_t}(1 + r_{t-j}^s) - 1 \tag{18}$$

The first equation sets the lookback horizon as a function of the regime. The second compounds the past ℓ_t monthly returns into a stock score. Stocks are then ranked by this score. The top decile forms the long leg and the bottom decile the short leg, as described in Appendix C.1.

When $\pi_t^{\text{filter}} \approx 0$ (calm), $\ell_t = 12$, recovering the standard 12-month momentum signal that captures return continuation (Jegadeesh and Titman, 1993). When $\pi_t^{\text{filter}} \approx 1$ (panic), $\ell_t = 1$, switching to a 1-month signal that exploits the short-horizon reversal documented in Jegadeesh (1990) and Lehmann (1990). At intermediate values, the lookback smoothly interpolates between these extremes.

LR: Logistic Regression

A logistic regression classifier is trained to predict whether a stock's forward return exceeds the cross-sectional median in each month. Classification is chosen over direct return regression because monthly stock returns are extremely noisy, and predicting the binary outcome "above or below median" is a more stable learning target that still produces a useful ranking of stocks. Continuous return regression remains the dominant target in the cross-sectional ML asset-pricing literature (Gu et al., 2020). Classification is used here as a deliberately simpler, lower-variance alternative for the linear baseline.[10]

$$P(y_t^s = 1 \mid \mathbf{x}_t^s) = \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_t^s) \tag{19}$$

where $y_t^s = \mathbb{1}(r_{t+1}^s > \text{median}_s\{r_{t+1}^s\})$ is a binary label indicating whether stock s outperforms the cross-sectional median in the following month. The feature vector $\tilde{\mathbf{x}}_t^s = (mom_{1,t}^s, \ldots, mom_{12,t}^s, \pi_t^{\text{filter}})$ contains the 12 momentum lookbacks and the regime signal, standardized using training-set means and standard deviations. The model is

[9] This is in the spirit of Goulding et al. (2023), with two simplifications: a continuous panic probability replaces their discrete four-state cycle classification, and a single regime-dependent horizon replaces their linear blend of one- and twelve-month signals.

[10] The logistic sigmoid function $\sigma(z) = 1/(1 + e^{-z})$ maps the linear combination $\boldsymbol{\beta}^\top \tilde{\mathbf{x}}$ from the real line to the interval $(0, 1)$, so the output can be interpreted as a probability.

estimated by maximizing the ℓ_2-regularized log-likelihood:

$$\hat{\boldsymbol{\beta}} = \arg\max_{\boldsymbol{\beta}} \left[\sum_i \Big(y_i \log \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i) + (1 - y_i) \log(1 - \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i)) \Big) - \frac{1}{2C} \|\boldsymbol{\beta}\|_2^2 \right] \tag{20}$$

with regularization strength $C = 1.0$. The ℓ_2 penalty shrinks coefficients toward zero to guard against overfitting given the large number of correlated momentum features. Once $\hat{\boldsymbol{\beta}}$ is estimated on the training sample, the stock score for any observation is obtained by plugging $\hat{\boldsymbol{\beta}}$ back into the prediction equation:

$$score_t^s = \sigma(\hat{\boldsymbol{\beta}}^\top \tilde{\mathbf{x}}_t^s) = P(y_t^s = 1 \mid \mathbf{x}_t^s;\, \hat{\boldsymbol{\beta}}) \tag{21}$$

Stocks with higher predicted probability of beating the cross-sectional median receive higher scores and are more likely to enter the long leg. Those with the lowest scores enter the short leg.

XGB: XGBoost Regressor

An XGBoost gradient-boosted tree regressor (Chen and Guestrin, 2016) is trained to predict the forward monthly return directly. Unlike LR, which classifies stocks as above or below the median, XGBoost predicts the forward return directly. This preserves magnitude information. A stock expected to gain 10% ranks higher than one expected to gain 2%, a distinction that binary classification discards. The feature vector for each stock-month observation comprises the 13 features described in Section 4.4: the 12 momentum lookbacks ($mom_1, \ldots, mom_{12}$) and the regime signal (π_t^{filter}). By including all horizons jointly, XGBoost can learn how the full momentum term structure interacts with the regime signal, for example by conditioning its joint use of all 12 horizons on the level of π_t^{filter} in ways that cannot be decomposed into per-horizon slope changes.

XGBoost builds an additive ensemble of M regression trees, where each successive tree fits the residuals of the current ensemble:

$$F(\mathbf{x}) = \sum_{m=1}^{M} \eta \cdot h_m(\mathbf{x}) \tag{22}$$

where h_m is the m-th decision tree and η is the learning rate that shrinks each tree's contribution to prevent overfitting. Each tree h_m is grown to minimize a regularized objective combining the squared-error loss on the current residuals with penalty terms on tree complexity (number of leaves and leaf weights). The predicted return $\hat{r}_{t+1}^s = F(\mathbf{x}_t^s)$ serves as the stock score. The model is trained with $M = 500$ trees, a learning rate of $\eta = 0.05$, maximum tree depth of 4, and row and column subsampling rates of 0.8. To reduce sensitivity to random initialization, an ensemble of 50 independent XGBoost models (each with a different random seed) is trained. Their predictions are averaged before ranking stocks into portfolio deciles.

The maximum depth was selected via cross-validation on the 1990–2010 training sample (Appendix G.3). Sharpe rises monotonically from 0.53 at depth 1 to 1.11 at depth 4 and declines beyond depth 5. Depth 4 is the budget required for the $K = 4$ cluster decomposition in Section 5.2.2, namely depth 1 for the regime gate plus depth 3 for within-regime shape reading.

XGBoost is chosen because its sequential residual fitting concentrates splits on whichever feature most reduces error, turning the regime signal into the model's primary gating variable rather than one feature among thirteen.

With the two-stage framework in place, the next section describes the data, sample construction, and the feature selection procedure used to choose the HMM's input features.

4 Data

This chapter introduces the inputs to the two-stage framework. The Stage 2 inputs are minimal by design: twelve trailing monthly returns paired with the regime signal (Section 4.4), so that the Section 5 mechanism story is not confounded with feature engineering. The discretionary choice lies in Stage 1: which four of nine candidate stress indicators the HMM should see. Section 4.1 introduces the four winners (DD, DISP, REL_N, CS), and Section 4.2 documents the four-pass selection procedure that picked them, designed so each pass shrinks the candidate set on a different criterion (in-sample regime quality, downstream portfolio Sharpe, validation at higher seed counts, cross-seed stability) rather than on a single in-sample fit. Section 4.3 reports the resulting regime signal π_t^{filter} and confirms it tracks the dot-com bust, GFC, COVID, and 2022 episodes. Section 4.5 reruns the entire selection procedure on UK and Japanese equities and finds that different regions select different feature sets, demonstrating that the chapter's contribution is a transferable procedure rather than a US-specific feature artefact.

The stock-level sample spans January 1990 to November 2025 and includes all ordinary common shares (CRSP share codes 10 and 11) listed on the NYSE, AMEX, or NASDAQ with a price above \$1.[11] Following Shumway (1997), delisting returns are incorporated to prevent survivorship bias: actual delisting returns are used when available, and −30% is imputed for performance-related delistings with missing returns.[12] The sample is split into a training period (1990–2010) used to estimate all models and a test period (2011–2025, 167 months) used exclusively for out-of-sample evaluation. No model parameters are re-estimated during the test period. Table 1 provides an overview.

	Train (1990–2010)	Test (2011–2025)	Full
Unique stocks	13,363	7,037	16,657
Stock-month observations	1,161,014	557,087	1,718,101
Avg. stocks per month	4,838	3,336	4,221
Market-level months	240	167	407

Table 1: Sample summary. The sample includes all ordinary common shares listed on NYSE, AMEX, or NASDAQ with price above \$1.

4.1 HMM Input Features

The HMM classifies market regimes using four monthly macrofinancial stress indicators. Each captures a distinct dimension of market stress:

[11] The \$1 price floor admits small and illiquid stocks into the universe, but NYSE decile breakpoints and value-weighting within each leg attenuate their influence on the realised portfolio.

[12] Full data sources and detailed variable definitions are provided in Appendix A.

1. **DD (Drawdown):** The percentage decline of the cumulative market price index from its rolling 12-month peak:

$$DD_t = \frac{P_t - \max(P_{t-11}, \ldots, P_t)}{\max(P_{t-11}, \ldots, P_t)} \tag{23}$$

DD directly measures the depth of ongoing market declines. Daniel and Moskowitz (2016) show that momentum crashes are predictable from past market declines, making DD the natural anchor for regime identification.

2. **DISP (Return Dispersion):** The log cross-sectional standard deviation of individual stock returns:

$$DISP_t = \log(\text{std}_s(r_{s,t})) \tag{24}$$

DISP captures cross-sectional *disagreement* among stocks. During stress, individual stocks move in different directions rather than together, causing dispersion to spike.

3. **REL_N (Relative Market Participation):** The log ratio of the number of actively traded stocks to its trailing 12-month average:

$$REL\_N_t = \log\left(\frac{N_t}{\frac{1}{12}\sum_{\ell=1}^{12} N_{t-\ell}}\right) \tag{25}$$

REL_N turns negative during crises when stocks delist, halt trading, or are acquired at elevated rates. It captures structural market damage that the other features do not measure.

4. **CS (Credit Spread):** The difference between Moody's BAA and AAA corporate bond yields:

$$CS_t = Y_t^{\text{BAA}} - Y_t^{\text{AAA}} \tag{26}$$

The credit spread provides an independent signal from the corporate bond market, helping the HMM distinguish genuine stress episodes from equity-specific turbulence.

All four features are winsorised at the 1st and 99th percentiles[13] using training-sample statistics, and standardised to zero mean and unit variance.

[13] Winsorising at the 1st and 99th percentiles caps any value below the 1st percentile at that bound and any value above the 99th percentile at that bound, limiting the influence of outliers without discarding the observations.

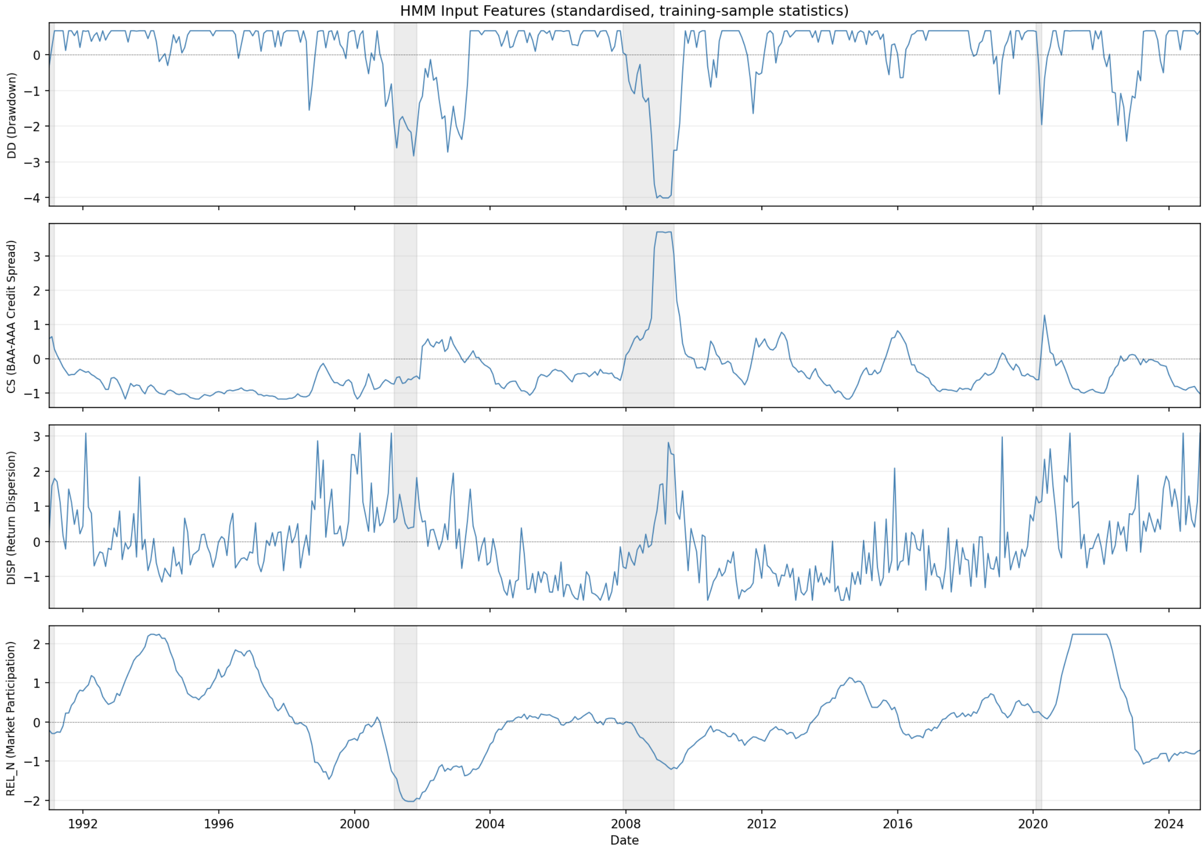


Figure 3: Time series of the four HMM input features (DD, DISP, REL_N, CS), standardised using training-sample statistics. Shaded regions indicate NBER recession periods.

The four series co-move clearly around stress episodes: DD and REL_N drop while DISP and CS spike during the 2000–2002 dot-com bust, the 2007–2009 Global Financial Crisis, the March 2020 COVID shock, and the 2022 Fed-hiking cycle, with the shaded NBER recession bands aligning with the largest of these joint excursions. This co-movement is what makes a single hidden-state variable viable in the HMM: the four features carry overlapping information about market stress, so one regime label can summarise the multivariate pattern.

4.2 Feature Selection

The regime signal's ability to capture the shift in momentum's cross-sectional structure depends critically on which stress indicators the HMM observes. The selection problem has two constraints. First, market drawdown (DD) must be included: it is the most economically direct measure of the mechanism identified by Daniel and Moskowitz (2016) (momentum crashes follow market declines) and consistently emerges as the dominant regime-identifying feature. Second, the number of features is capped at four. With 241 training months, each regime has sufficient observations to estimate a 4-dimensional mean vector and covariance matrix (10 free covariance parameters per state). Increasing beyond four features risks estimation instability.

Candidate pool

Nine monthly macrofinancial indicators, all available from 1990 onward, are considered:

- *Equity market:* market drawdown (DD), realised volatility (VOL), cross-sectional return dispersion (DISP), relative market participation (REL_N), advance-decline ratio (ADR), and cross-sectional return skewness (SKEW).
- *Credit and macro:* BAA–AAA credit spread (CS), log VIX (LVIX), and the 10-year minus 2-year Treasury term spread (TERM).

All candidates are observable in real time at the end of each month and standardised to z-scores using training-period statistics.

Selection procedure

The selection proceeds through four passes of increasing rigour. In the first pass (*quality screen*), all DD-inclusive combinations of up to four features are considered. With 8 remaining candidates, this gives $\binom{8}{0} + \binom{8}{1} + \binom{8}{2} + \binom{8}{3} = 1 + 8 + 28 + 56 = 93$ combinations (DD alone, DD paired with one other, DD with two others, DD with three others). Each is estimated with a single Gibbs sampler seed, and combinations are eliminated if the HMM fails basic sanity checks: the two states must separate meaningfully, and the model must correctly classify known crises (the dot-com bust, the GFC, and COVID) as panic while keeping non-crisis periods calm.[14]

In the second pass (*portfolio ranking*), surviving combinations are evaluated through the full cross-sectional pipeline: HMM to XGBoost to long-short portfolio. This answers the question that matters most: does a better regime signal translate into better stock selection? Combinations are ranked by out-of-sample Sharpe ratio at the selection seed budget (10 HMM seeds, 20 XGBoost seeds). The top 10 advance.

In the third pass (*definitive validation*), the surviving finalists are re-evaluated at higher seed counts to confirm that the Pass-2 ranking is not an artefact of low-seed MCMC noise.

In the fourth pass (*stability check*), the finalists are tested for robustness across random seed partitions. A combination that achieves a high Sharpe on one lucky seed combination but collapses on another is unreliable. We run 20 independent experiments with different seed draws and classify combinations as stable ($\sigma < 0.10$), moderate ($0.10 \leq \sigma < 0.15$), or unstable ($\sigma \geq 0.15$).

[14] Specific thresholds: ESS > 50, regime separation significant for at least $\min(2, D)$ features, average $\pi_t^{\text{filter}} > 0.5$ during GFC and COVID, < 0.4 during 2003–2006 and 2013–2019, > 0.3 during the dot-com crash.

Selected features

The four-pass procedure converges on DD, DISP, REL_N, and CS. The choice is driven by complementarity rather than peak Sharpe. DD is the most economically direct measure of stress and the strongest single-feature classifier (DD-only Sharpe 0.66 vs full-set 0.97 in Table 13, Appendix B). Leave-one-out experiments in the same table show that REL_N and DD are the two largest individual contributors to the full-set Sharpe (drops of −0.54 and −0.46 respectively when each is removed), confirming that REL_N captures structural damage (delistings, trading halts, forced acquisitions) that drawdown alone misses. DISP and CS contribute marginally on a leave-one-out basis (−0.01 each in Sharpe). The full feature set has materially lower cross-partition σ (0.11) than DD alone (0.14) (Table 12), and DISP (cross-sectional disagreement among stocks) and CS (corporate credit conditions) carry stress information that is independent of price drawdown and contributes to this stability.

The four-feature combination ranks first on both single-experiment ensemble Sharpe and cross-partition mean Sharpe in Table 12. Sharpe ratios in that table are based on the selection pipeline's seed budget (10 HMM seeds, 20 XGBoost seeds), kept small so the procedure can search many candidate combinations within a reasonable compute budget. The production configuration runs the chosen combination at a much larger seed budget (200 HMM seeds, 50 XGBoost seeds), which raises the Sharpe to 1.11 (Section 5). The seed-convergence experiment in Appendix J.2 confirms that this lift is a saturation effect: averaging more Gibbs chains produces a cleaner regime signal π_t^{filter}, and the resulting Sharpe rises monotonically with the number of averaged chains, plateauing near 1.12 from $k = 30$ chains onwards. The 200-chain production setting sits at that plateau.[15]

4.3 Regime Estimation Results

The convergence diagnostics described in Section 3.1.5 confirm that the Gibbs sampler with the selected features has converged. All posterior parameters exceed an ESS of 100,[16] meaning the post-burn-in draws contain at least 100 effectively independent samples per parameter. The Gelman–Rubin $\hat{R}$ statistic is below 1.01 for every parameter across five independent chains (Appendix J.1),[17] indicating that chains started from different

[15] Pass 2 evaluates feature sets on test-period Sharpe, so the selected specification partly reflects test outcomes. The Pass 4 final choice is made on cross-seed stability, and Tables 12 and 13 show that all $D = 1$ to $D = 4$ counts and leave-one-out variants produce Sharpes from 0.57 to 0.91. The headline 1.11 should be read as the upper end of a tight range, not a point isolated by search.

[16] The effective sample size (ESS) is the number of effectively independent draws an autocorrelated MCMC chain is worth: n draws yield ESS $\leq n$, with equality only if the draws are independent. ESS > 100 per parameter is a standard minimum for reliable posterior inference (Gelman et al., 2013). Formal definition in Section 3.1.5.

[17] The Gelman–Rubin $\hat{R}$ compares within-chain and between-chain variance across multiple independent chains. Values near 1 indicate that chains started from different initialisations have settled at the same posterior, with $\hat{R} < 1.1$ the standard threshold (Gelman and Rubin, 1992). Formal definition in Section 3.1.5.

initializations have converged to the same posterior distribution.

Table 2 and Figure 4 summarize the HMM output. All four features exhibit statistically significant separation between calm and panic regimes (95% credible intervals excluding zero), and the resulting signal correctly identifies all major stress episodes while remaining near zero during calm expansions.

	$\hat{\mu}_{\text{calm}}$	$\hat{\mu}_{\text{panic}}$	Δ	95% CI
DD_z	0.531	-0.976	-1.507	$[-1.794, -1.215]$
DISP_z	-0.423	$+0.689$	$+1.113$	$[+0.869, +1.353]$
REL_N_z	0.525	-0.895	-1.419	$[-1.591, -1.231]$
CS_z	-0.697	$+0.203$	$+0.900$	$[+0.626, +1.159]$

Table 2: HMM posterior regime-mean separation.

Notes: This table reports posterior mean estimates of regime-conditional feature means from the Bayesian two-state HMM, estimated via Gibbs sampling on the 1990–2010 training period. All features are standardised to zero mean and unit variance. $\Delta = \hat{\mu}_{\text{panic}} - \hat{\mu}_{\text{calm}}$ is the separation between regimes. The 95% CI is the Bayesian posterior credible interval for Δ; zero lies outside all intervals, confirming significant regime separation. DD = market drawdown; DISP = log cross-sectional return dispersion; REL_N = relative market participation; CS = credit spread (BAA−AAA).

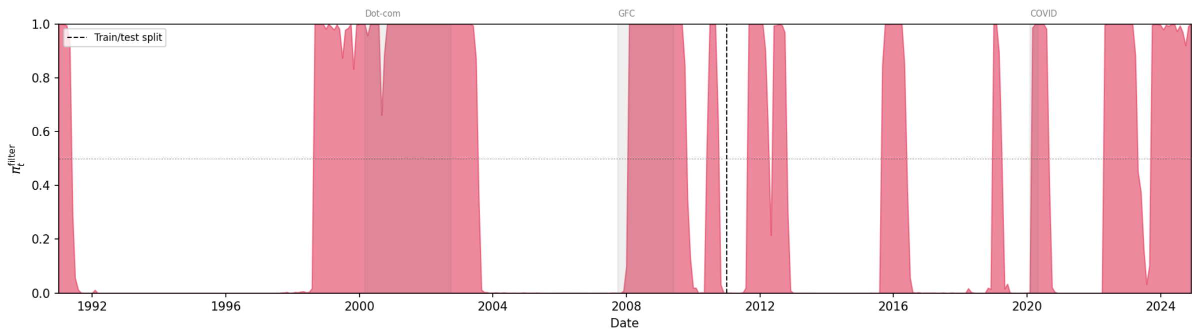


Figure 4: Filtered regime probability π_t^{filter} (1990–2025). The pink area shows the filtered probability path. The dashed vertical line marks the train/test split (January 2011), the dotted horizontal line marks the 0.5 threshold above which a month is classified as panic, and the grey vertical bands mark NBER recession periods. Major stress episodes (Dot-com, GFC, COVID) are labelled at the top.

4.4 Cross-Sectional Features

The cross-sectional models rank stocks each month using momentum signals and the filtered panic probability π_t^{filter}.

Momentum lookbacks

Twelve trailing momentum signals are computed for each stock, corresponding to lookback horizons of 1 to 12 months:

$$\mathrm{mom}^s_{k,t} = \prod_{\ell=1}^{k}(1 + r^s_{t-\ell}) - 1, \qquad k = 1, \ldots, 12 \tag{27}$$

where $r^s_{t-\ell}$ is the adjusted return of stock s in month $t - \ell$. The current month's return is excluded from all lookbacks to avoid mechanical correlation with the forward return being predicted. Including all twelve horizons rather than a single lookback allows the models to learn how the full momentum term structure interacts with the regime signal.

Regime signal

The filtered panic probability π^{filter}_t from the HMM enters as a single feature. Because it varies only at the monthly market level (every stock in the same month receives the same value), it functions as a conditioning variable rather than a stock-level predictor.

4.5 International Sample

For the robustness check in Section 5.3.4, the same pipeline is applied to UK and Japanese equities from Compustat Global over the same 1990–2025 window. The HMM feature set is re-selected per region using the four-pass procedure described in Section 4.2, run separately on each market's data. The candidate pool substitutes `BANK_REL` (the value-weighted bank-sector excess return over the broad regional market, sign-flipped so high values indicate stress) for the US-specific Moody's BAA–AAA spread, and both regions select *DD+VOL+REL_N*. Long and short legs are formed from unconditional decile breakpoints across all listed stocks, since no NYSE-equivalent tier exists for either market. Per-region selection details and stability checks are in Appendix K.

5 Results

This chapter presents the empirical evidence that answers the thesis's central research question: how does the cross-sectional structure of momentum change across regimes, and what mechanism drives that change? Three sections build the answer in turn. Section 5.1 reports the out-of-sample performance of the three portfolio-construction methods developed in Section 3 (DET, LR, XGB) against unconditional momentum benchmarks, establishes XGB as the only variant that captures the regime-momentum signal, and ablates that signal to confirm its essential role. Section 5.2 opens up the chosen model month by month with SHAP decomposition and a cross-sectional z-score clustering, identifying four recurring selection modes that constitute the empirical mechanism. Section 5.3 tests the robustness of the result to a risk-return trade-off, a 25-year expanding-window back-test covering the dot-com bust and the Global Financial Crisis, a stress test characterising the strategy's reversal-failure vulnerability, and an international replication on UK and Japanese equities. The mechanism findings synthesised here form the basis of the contributions discussed in Section 6. All strategies are evaluated on 2011–2025 (167 months) using the long-short construction defined in Section 3.2. Data and feature definitions are in Section 4.

5.1 Portfolio Performance

5.1.1 Results Overview

Table 3 reports the out-of-sample performance of all strategies.[18]

	Ann. Ret	Ann. Vol	Sharpe	Max DD	β	NW t	Final \$
Market	11.9%	14.6%	0.84	-24.7%	1.00	–	4.8
Fixed 12-mo mom	-1.9%	26.1%	0.06	-72.2%	-0.66	0.24	0.8
Fixed 1-mo mom	3.1%	17.6%	0.26	-30.2%	-0.34	1.08	1.5
DET	-0.2%	23.6%	0.11	-63.8%	-0.59	0.43	1.0
LR	-1.3%	15.2%	-0.01	-42.4%	0.05	-0.04	0.8
XGB	21.7%	19.5%	1.11	-22.8%	0.45	4.37***	15.3

Table 3: Out-of-sample portfolio performance.

Notes: Long-short (top/bottom decile, NYSE breakpoints), value-weighted, net of 10 bps one-way costs. β is the market beta. NW t uses Newey–West standard errors (6 lags). Test period: January 2011 to November 2025 (167 months). $^{*}\,p < 0.10$; $^{**}\,p < 0.05$; $^{***}\,p < 0.01$.

Table 3 reports out-of-sample performance for the three cross-sectional model variants (DET, LR, XGB) alongside three benchmarks: the value-weighted U.S. market and two

[18] Sharpe convention details (raw vs. excess returns, arithmetic-vs-geometric mean) are in Appendix C.3.

unconditional momentum portfolios at the dominant 12-month and contrarian 1-month horizons. The model variants share the same inputs, the twelve monthly momentum horizons together with the HMM filtered probability π_t^{filter}, but differ in functional form: DET is a deterministic rule that interpolates between mom_1 and mom_{12} as π_t^{filter} varies; LR is a logistic regression on those features; XGB is an XGBoost regressor capable of representing arbitrary nonlinear interactions.

The pattern across rows is clear. The value-weighted market earns 11.9% annualised at Sharpe 0.84, reported as the passive equity benchmark.[19] Unconditional 12-month momentum (the canonical anomaly) delivers only Sharpe 0.06 with a −72.2% maximum drawdown, exposing the crash vulnerability documented by Daniel and Moskowitz (2016) and Barroso and Santa-Clara (2015). The contrarian 1-month version reaches 0.26. Of the regime-conditional variants, the deterministic rule (DET) improves on 12-month momentum only marginally to 0.11, and the logistic regression (LR), given identical features to XGB, collapses to −0.01. Only XGB delivers economically meaningful performance: an annualised Sharpe of 1.11 with a six-factor alpha of 24.1% ($t = 4.81$; Appendix H.3), the only row in Table 3 whose Newey-West t-statistic clears every conventional threshold. Block bootstrap paired tests reject equality of Sharpe ratios between XGB and each benchmark at $p < 0.025$; XGB's own 95% Sharpe confidence interval is $[0.67, 1.53]$, comfortably above zero, while every benchmark's interval straddles zero (Appendix H.4). An expanding-window backtest extending the OOS sample to 25 years (2000–2024) reproduces this Sharpe within sampling noise on the 167-month overlap with the main test period (0.88; see Section 5.3.2, which also covers the dot-com bust and GFC out-of-sample).

A further specification confirms that the binding constraint is linearity itself. Hand-engineering the missing nonlinearity into LR through π^2 and $\pi \times mom_k$ terms lifts it to 0.32, establishing that the regime-momentum interaction is real while also showing that pre-specified interactions recover less than a third of what XGBoost finds automatically. Adding firm-level fundamentals to XGB's feature set degrades performance rather than improving it (Section 5.1.3), evidence that π_t^{filter}'s role is specifically to condition the momentum term structure. XGB is therefore adopted as the production model for the remainder of the thesis. The following sections analyse its decision rule (Section 5.2), test its robustness (Section 5.3), and replicate the construction internationally (Section 5.3.4).

[19]The market is long-only with full market exposure ($\beta = 1$), while every other row in Table 3 is long-short and approximately market-neutral, so raw Sharpe ratios are not directly comparable across the long-only/long-short divide: the market harvests the equity premium, whereas the long-short rows earn their return from cross-sectional ranking. The like-for-like Sharpe peers for the model variants are therefore the unconditional long-short momentum benchmarks. The proper market-relative test that controls for exposure is the six-factor alpha (Appendix H.3).

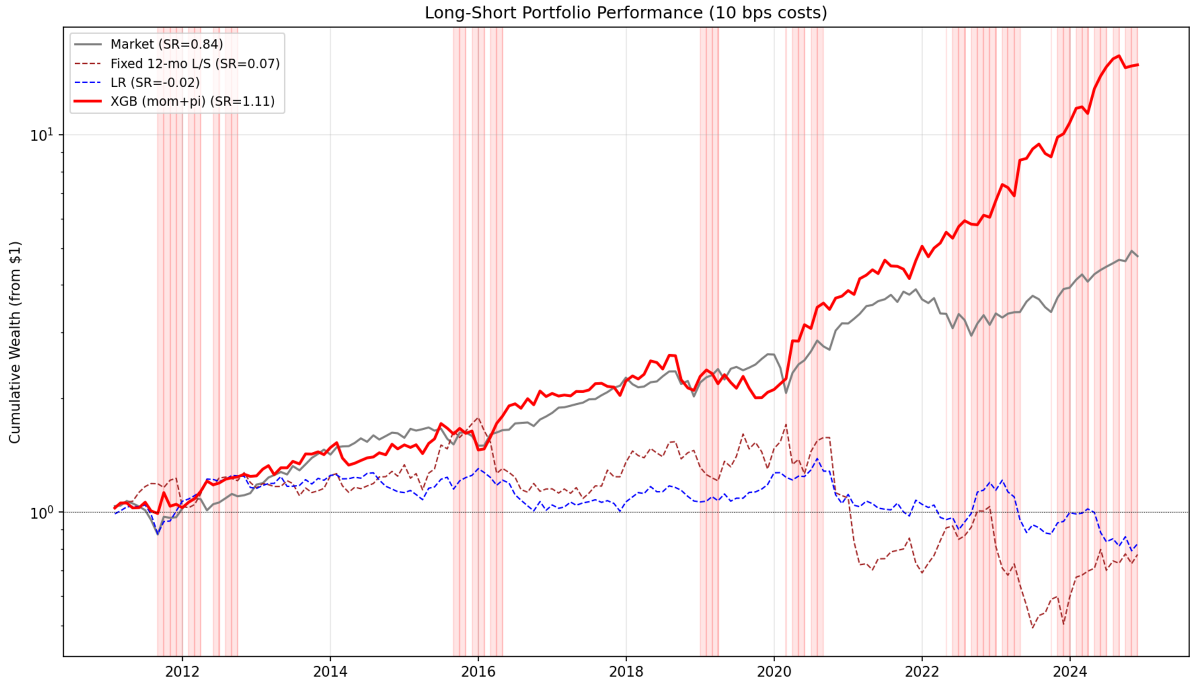


Figure 5: Cumulative wealth paths (log scale) starting from \$1 in January 2011. Shaded bands indicate panic months ($\pi_t^{\text{filter}} \geq 0.5$).

The regime-conditional view (Table 4) shows where this advantage comes from. XGB's panic Sharpe (1.53) is roughly double its calm Sharpe (0.84). The point-estimate gap is consistent with the regime-conditioning mechanism, but the block-bootstrap paired test does not reject equality at the 5% level (Appendix H.4, $p = 0.109$). Statistical confirmation would require a longer panic sample. Every benchmark either collapses (LR: 0.39) or stays near zero (Fixed 12-mo: 0.05) in panic. The advantage is concentrated in the regime where unconditional momentum is most exposed.

	Full	Calm	Panic
Market	0.84	0.64	1.13
Fixed 12-mo mom	0.06	0.08	0.05
Fixed 1-mo mom	0.26	0.52	−0.01
DET	0.11	0.18	−0.01
LR	−0.01	−0.29	0.39
XGB	1.11	0.84	1.53

Table 4: Regime-conditional Sharpe ratios (108 calm months, 59 panic months).

5.1.2 Regime Signal Ablation

The regime signal is essential. Removing π_t^{filter} and retraining on the 12 momentum features alone drops the Sharpe to 0.43 (Table 5). Replacing π_t^{filter} with the four raw stress indicators (DD, DISP, REL_N, CS) performs even worse (Sharpe 0.26). With four additional dimensions the model must reconstruct the regime structure at each split, while the HMM distils those indicators into a single dimension that one tree split can condition on. The HMM's role is not to provide a convenient feature but to solve regime identification *before* stock selection, freeing the cross-sectional model to use the regime signal rather than construct it. If the model cannot extract a meaningful regime signal even when given the four raw indicators directly, it certainly cannot do so from momentum features alone, which are one further step removed from the underlying regime dynamics.

Regime Signal Variant	Ann. Ret	Ann. Vol	Sharpe	Max DD
XGB + π_t^{filter} (HMM)	21.7%	19.5%	1.107	−22.8%
XGB + raw indicators (DD, DISP, REL_N, CS)	3.2%	19.1%	0.257	−45.4%
XGB (no regime signal)	7.0%	20.4%	0.429	−41.0%

Table 5: Regime signal ablation (50-seed ensemble). The HMM signal outperforms both raw stress indicators and no regime signal.

5.1.3 Fundamental Feature Augmentation

The regime signal ablation removes π_t^{filter} from the production model. The complementary question is whether *adding* widely used firm-level fundamentals strengthens the strategy. Ten characteristics are tested,[20] each trained at production settings.

	Feat.	Sharpe	Calm	Panic	β	NW t (exc.)	Fund SHAP
Baseline (mom + π)	13	1.11	0.84	1.53	0.45	1.83*	0%
+ 10 fundamentals	23	0.89	0.69	1.18	0.07	0.61	44%
Fundamentals only	10	0.59	0.86	0.38	0.19	−0.49	100%
Fundamentals + π	11	0.64	0.59	0.72	0.03	−0.52	60%
Momentum + fund. (no π)	22	0.63	0.68	0.57	0.16	−0.29	54%

Table 6: Feature-set ablation (50-seed XGBoost ensembles, production hyperparameters). Full Sharpe, calm-period Sharpe, and panic-period Sharpe are reported alongside market beta and the Newey-West t-statistic on excess returns over the market (6 lags), matching the convention of Table 3. "Fund SHAP" is the share of total SHAP attention absorbed by the ten fundamental features.

Adding the fundamentals to the baseline reduces Sharpe from 1.11 to 0.89 (Table 6). The new features absorb 44% of total SHAP, redirecting attention away from the momentum term structure (54% to 27%) and the regime signal (46% to 29%) without net improvement. Beta falls from 0.45 to 0.07 and maximum drawdown tightens from −22.8%

[20] Book-to-market, return on equity, gross profitability, asset growth, leverage, earnings growth, log market equity, operating cash flow over assets, free cash flow over assets, and accruals. All real-time and four-month-lagged from Compustat `fundq`, with full definitions in Appendix L.

to −18.3%, so the augmented variant behaves more like a low-beta market-neutral strategy than a high-alpha regime-conditioned one.

The fundamentals-only variant identifies the source of the dilution. Its calm Sharpe of 0.86 essentially matches the baseline's 0.84, but its panic Sharpe of 0.38 is a fraction of the baseline's 1.53: stable firm attributes do not flip direction when the market enters panic. Adding the regime signal to the fundamentals (fundamentals $+\pi$) lifts panic Sharpe to 0.72 but remains well short of the full model, because without the momentum term structure there is no directional relationship for the regime signal to reorganise. The regime-momentum interaction is the load-bearing mechanism. Fundamentals cannot substitute for it. A regime-conditional feature-set design is discussed in Section 6.2.

5.2 The Regime-Momentum Mechanism

This section opens the XGBoost black box, reading the structure of the regime-momentum signal directly off the model's per-month selections.

Section 5.2.1 uses SHAP to establish that both π_t^{filter} and the momentum horizons are decision-relevant; Section 5.2.2 traces *what shape* the model picks via cross-sectional z-scores and clusters those z-curves into discrete selection modes; Section 5.2.3 synthesises these patterns into a regime-conditional mechanism with two interlocking facets and contrasts them with the prior crash-management literature.

5.2.1 Per-Horizon Momentum SHAP

XGB's predictions are decomposed using SHAP values (Lundberg and Lee, 2017), which measure how much each input contributes to the model's prediction (definitions in Appendix D.1). Aggregate feature importance is split almost evenly. π_t^{filter} accounts for 46% of total SHAP, the 12 momentum horizons together for 54% (Table 7). The near-even split supports a routing-variable role for π_t^{filter}. It determines how the model uses the momentum term structure. Between regimes, momentum's share rises from 51% in calm to 59% in panic while π_t^{filter}'s share falls from 49% to 41%.

Feature	Overall	Calm	Panic
Momentum (12 horizons)	54%	51%	59%
π^{filter}	46%	49%	41%

Table 7: SHAP feature importance shares for XGB. Each column sums to 100%.

The aggregate split is reported by leg and horizon in Figure 6. Both legs concentrate on months 8–12 (about 65% each), with months 11 and 12 carrying the largest individual shares. The short leg's peak at month 11 (22%) is the most concentrated single-horizon share in the figure.

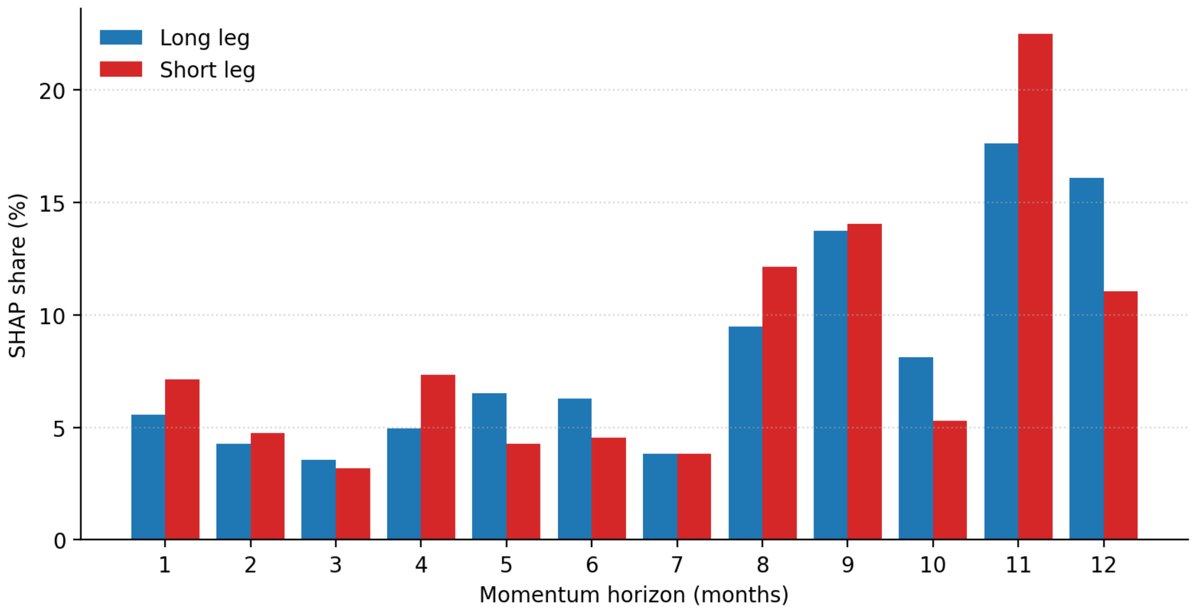


Figure 6: Per-horizon SHAP shares of momentum features in XGBoost's predictions, by leg, pooled across calm and panic regimes weighted by stock-month counts.

SHAP shows which horizons matter, but not in which direction. The same SHAP weight could reflect **continuation** (high momentum predicts high return, the classical Jegadeesh and Titman (1993) result), **reversal** (high momentum predicts low return), or **state-dependent selection** (the sign of the relationship flips with the regime). Only the third possibility requires a nonlinear model. A linear regression can encode constant continuation or constant reversal, but cannot encode a relationship whose sign depends on the regime. XGB selects winners in calm regimes and losers in panic regimes, the state-dependent selection case. The same momentum signal that the model rewards in calm is penalised in panic, so the regime signal flips the sign of the momentum-return relationship. Sections 5.2.2 and 5.2.3 trace this flip cluster by cluster.

5.2.2 Cross-Sectional Z-Scores

SHAP shares identify which inputs matter but not which stocks the model actually picks. To trace the picks themselves, we use cross-sectional z-scores. For each stock in the long or short leg, the z-score at horizon k is $(mom_k^s - m\bar{o}m_k)/\sigma_k$, computed from the cross-sectional mean and standard deviation of all stocks that month. Figure 7 plots the long-leg z-curve month by month.

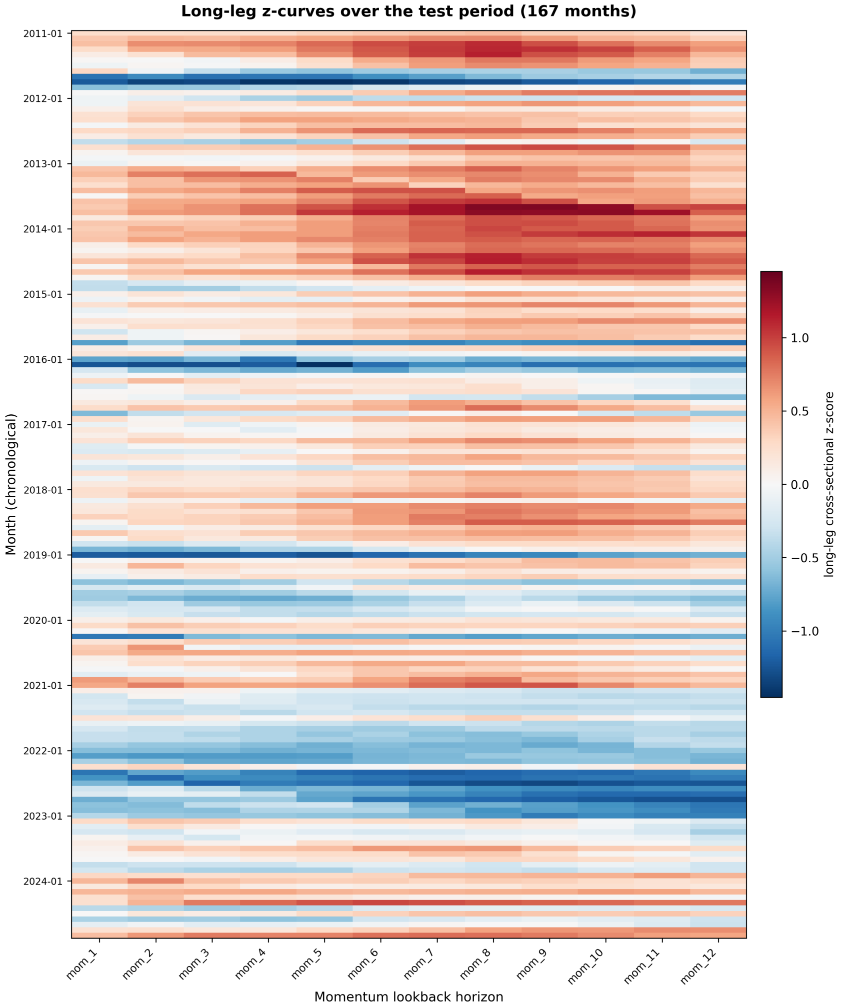


Figure 7: Long-leg cross-sectional z-scores by month and momentum horizon, 2011–2025. Each row is one of 167 test months sorted chronologically. Each column is one of the 12 momentum horizons. Color shows the deviation of the long leg's mean momentum at horizon h from the cross-section average that month.

The heatmap reveals a small number of recurring selection modes. To summarise these, we cluster the 167 monthly z-score profiles with K-means ($K = 4$, treating each month as a 12-dimensional point). $K = 4$ is chosen on economic grounds. We do not use $K = 3$ because the coarser partition collapses the two below-average reversal modes (post-panic recovery and deep-crisis term-structure-wide reversal) into a single cluster, eliminating a distinction central to the mechanism (Section 5.2.3). We do not use $K \geq 5$ because the four-cluster partition already covers the two above-average modes and two below-average modes that the mechanism turns on, and further splitting would partition an economically homogeneous group without revealing a new selection pattern. The partition is highly stable across random initialisations.[21] The four groups nest inside a

[21] The same labels emerge across six independent random seeds for 163 of 167 months (97.6%), with mean pairwise Adjusted Rand Index 0.96. Pairwise centroid distances in 12-dimensional z-space range from 1.37 (clusters 1 and 2, both above-average) to 4.90 (clusters 1 and 4, the extremes), against a typical within-cluster scatter of 0.6 to 0.9.

coarser two-fold partition. Clusters 1 and 2 are the 115 months in which the long leg sits above the cross-section average ("above-average baskets", $\bar{\pi} = 0.28$). Clusters 3 and 4 are the 52 months below ("below-average baskets", $\bar{\pi} = 0.52$).

Per-cluster descriptive statistics are in Table 8. The full feature signature is in Appendix Table 26. Each cluster's z-curve panel appears immediately above its description (Figures 8–11). Thin lines are member-month z-curves and the bold line is the cluster centroid.

Feature	1 (calm)	2 (transition)	3 (post-panic)	4 (deep crisis)
Months (n)	53	62	31	21
$\bar{\pi}_t^{\text{filter}}$ (current panic prob.)	0.21	0.34	0.32	0.81
Cross-section std of 9–12 mo momentum	0.60	0.59	0.85	0.45
Cross-section skewness of 1–4 mo momentum	4.6	7.2	6.4	3.7
Long-leg avg z-score $\bar{z}$	+0.57	+0.19	−0.29	−0.82
Long-leg avg trailing return	+40%	+16%	−3%	−37%
Cross-section avg trailing return	+14%	+7%	+13%	−9%
Short-leg avg trailing return	−3%	+0%	+42%	−4%
Cluster Sharpe	0.92	0.93	1.21	1.82

Table 8: $K = 4$ cluster descriptors, 2011–2025 ($n = 167$ months); columns are clusters. The trailing-return rows (long, cross-section, short) are averaged across the 12 momentum horizons. Dispersion is shown at 9–12 months and skewness at 1–4 months, the windows where clusters differ most. Cluster Sharpe is annualised, block-bootstrapped (block size 6, 5,000 reps).

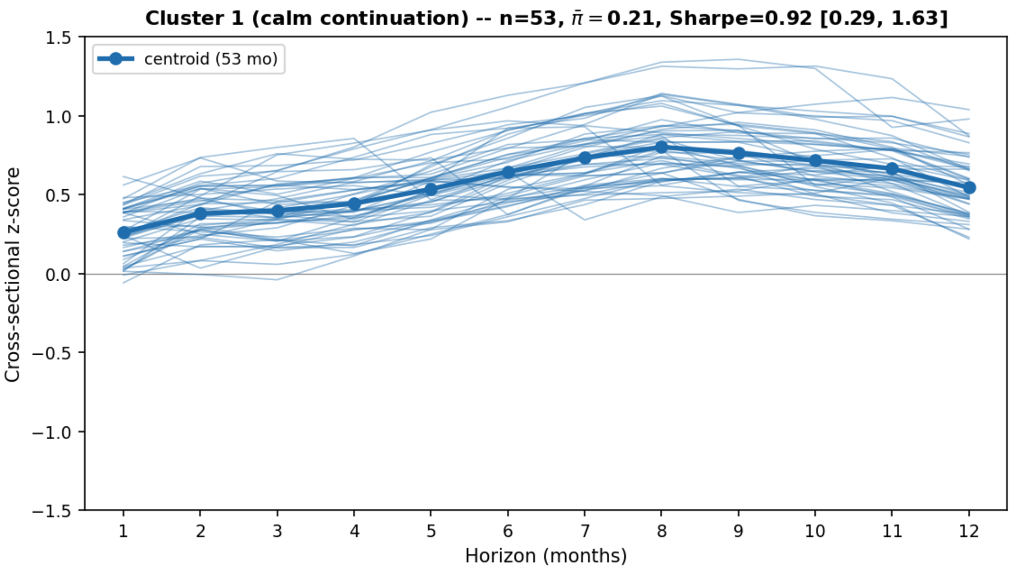


Figure 8: Cluster 1 (calm continuation): z-curves of the 53 member months and centroid. The centroid is humped and uniformly positive, peaking at +0.80 at month 8.

Cluster 1 (calm continuation): a quiet, broadly trending market

The long-leg z-curve has a z-mean of +0.57, the largest average amplitude on the continuation side, with a humped shape. Picks sit only mildly above average at short horizons ($z \approx +0.26$ at month 1) but are clearly differentiated at intermediate horizons (peak +0.80 at month 8). In absolute terms, raw pick momentum reaches +60% at month 9. The cluster is concentrated in 2013–2014, running in stretches of up to 21 consecutive months.

These 53 months are a stable, mature bull market. With cluster-mean $\bar{\pi}_t^{\text{filter}} = 0.21$ (the lowest of any cluster), the model has high confidence the regime is calm and commits fully to its standard winner-selection mode. The amplitude of the z-curve reflects that confidence. The cross-section is positive at every horizon (average stock +3%/+13%/+23% over 1/6/12 months), so there is a coherent up-trend for the model to ride. The rise is broad rather than driven by a few outliers (cross-section skewness 4.6/6.2/6.7, below the test-period average of 5.8/6.6/7.1), which means winner-selection picks up genuine widespread participation rather than chasing a handful of unrepresentative names. The cluster sits far from the post-bear-market rebounds where unconditional momentum is known to crash.

This is the canonical setting for cross-sectional momentum, the kind of extended bull market in which Cooper et al. (2004) document that momentum profits are most concentrated. The picks reproduce the buy-side of Jegadeesh and Titman (1993)'s twelve-month relative-strength strategy, with the intermediate-horizon peak matching Novy-Marx (2012)'s documented dominance. The underlying mechanism is Hong and Stein (1999)'s gradual information diffusion, in which news enters prices slowly so winners keep winning until the information is fully absorbed.

Cluster 1 earns a Sharpe of 0.92, the lowest of the four. The Sharpe is moderate because the trend is mature. Continuation still pays, but the strategy is riding an established rally rather than catching a turning point. Cluster 1 is a sanity check on the architecture, since at the lowest π_t^{filter} values in the test sample the model recovers the classical cross-sectional momentum pattern. The regime-conditioning architecture adds new behaviour during stress without overriding the unconditional pattern that already works in calm.

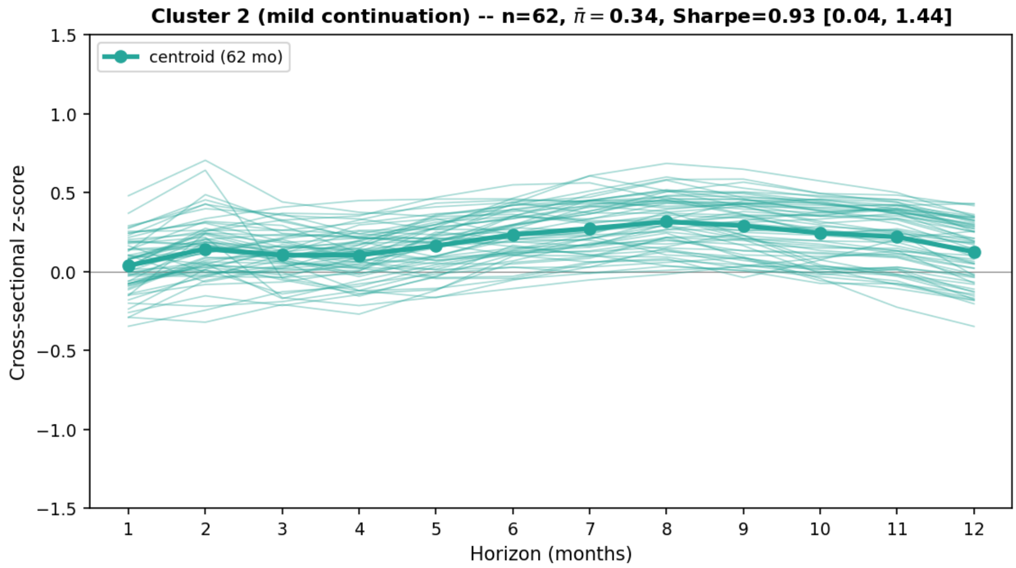


Figure 9: Cluster 2 (transition): z-curves of the 62 member months and centroid. The centroid is mildly positive throughout, peaking at +0.32 at month 8.

Cluster 2 (transition): a choppy market with no broad direction

The long-leg z-curve is nearly flat, with most horizons sitting between +0.10 and +0.30 (z-mean +0.19, peak +0.32 at month 8), the smallest absolute amplitude of any cluster, reflecting a hedged selection rather than a confident commitment. Cluster 2 is the largest of the four ($n = 62$) but appears only in short bursts (max 4 consecutive months) spread throughout the test period.

These 62 months are an ambiguous transition state, with cluster-mean $\bar{\pi}_t^{\text{filter}}$ in a middle band (0.34, neither clearly calm nor clearly panic). The cross-section's average momentum is mild ($mom_{\text{overall}} = 0.07$, below the test-period average of 0.08), so no clear up-trend has formed. The cross-section is only mildly positive (+2%/+8%/+11% over 1/6/12 months) and cross-section skewness is the highest of any cluster (7.2/7.6/8.1), with a small number of extreme winners pulling the mean above zero while the median stock's return stays near flat, so even the modest positive level is not broad-based. With neither regime call supported, the model plays safe and earns its return through occasional outlier capture rather than systematic selection. The short-burst pattern is consistent with cluster 2's role as a brief transition between regimes rather than a stable state. The muted shape is consistent with Cooper et al. (2004)'s finding that momentum profits concentrate in clearly bullish prior states, fading when the market state is ambiguous, and with Hong et al. (2000)'s slow-diffusion mechanism, in which momentum profits concentrate in stocks completing their delayed information adjustment rather than spreading across the cross-section.

Cluster 2 earns a Sharpe of 0.93, exposing the limit of the regime-momentum mechanism. When π_t^{filter} sits in the ambiguous middle band and the cross-section carries no coherent direction, the model cannot build a strongly-tilted long leg even with the regime signal available. The 0.93 Sharpe is essentially identical to cluster 1's 0.92 but reflects a different mechanism. Because the long-leg z-curve sits only marginally above the cross-section average (z-mean +0.19 across all twelve horizons), the picks are weakly differentiated from the broader universe. Combined with the high cross-section skewness (7.2/7.6/8.1 over 1/6/12 months, the highest of any cluster), the leg's positive return reflects occasional inclusion of right-tail outliers from the broader cross-section rather than a systematic selection edge.

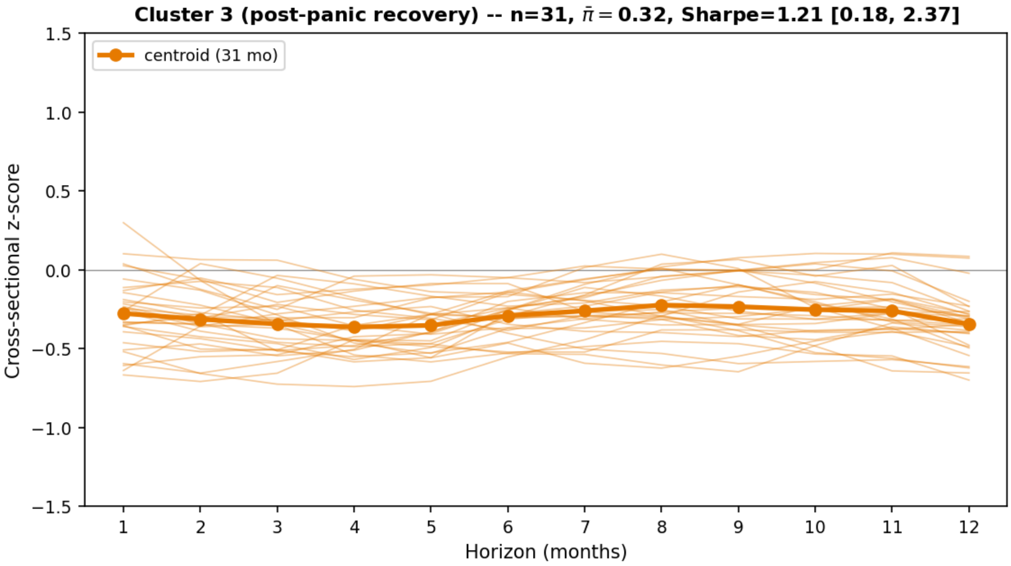


Figure 10: Cluster 3 (post-panic recovery): z-curves of the 31 member months and centroid. The centroid sits below the cross-section average at every horizon, between −0.22 and −0.36.

Cluster 3 (post-panic recovery): a recovering market with high long-horizon dispersion

The long-leg z-curve sits below the cross-section average at every horizon, with z-mean −0.29. The deepest below-average points are at month 4 (−0.36) and month 12 (−0.35). The amplitude is moderate, smaller than the extreme commitments seen in clusters 1 and 4. The picks' absolute trailing returns range from −6% to +1% while the broader cross-section is up double digits. The picks rotate more in this cluster than in any other. Only about 68% of the long-leg names carry over from one cluster-3 month to the next consecutive cluster-3 month (the lowest carry-over fraction of any cluster), as some recovery candidates catch up and others remain behind. Cluster 3 is concentrated in 2021 (9 of 31 months, the post-COVID rotation) and 2023.

These 31 months sit at a moderate cluster-mean panic probability ($\bar{\pi}_t^{\text{filter}} = 0.32$) paired with a strongly recovering cross-section (+1%/+11%/+27% over 1/6/12 months). The strong 12-month return paired with a near-flat most recent month tells us the rally is mature. The bulk of the rebound has already happened in the prior 6–12 months, and cluster 3 catches the late-stage decelerating phase. The joint signature of moderate π and strong positive cross-section average momentum routes the trees into a post-bear-rebound branch, where the long leg is built from cross-sectional losers rather than winners. Cross-section dispersion is the highest of any cluster at both the 5–8 and 9–12 month horizons (0.56 and 0.85), reflecting an uneven recovery in which some stocks rebound while others still lag. The increasing dispersion across horizons matches Lee and Swaminathan (2000)'s lifecycle interpretation, in which momentum signals evolve from continuation to reversal as information ages.

These months also exhibit the beta-reversion dynamic that Daniel and Moskowitz (2016) identify as crash-prone for unconditional momentum. During the preceding bear market, past losers were high-β stocks that fell disproportionately while past winners were low-β defensives that held up. When recovery arrives, the high-β losers rally disproportionately and the low-β defensives lag, inverting the cross-sectional ranking. Fixed 12-month momentum's long leg ends up holding lower-beta stocks than its short leg and gets dragged down when the high-β losers in the short leg rally; XGB flips this and rides the same beta-reversion that breaks fixed momentum (cluster-level CAPM betas in Section 5.2.3 below and Appendix H.7, Table 27).

Cluster 3 earns a Sharpe of 1.21 on these loser picks, in the same months where classical momentum is most vulnerable. What is distinctive about cluster 3 is that the model inverts to loser selection *while the cross-section itself is up*, a counter-intuitive bet that selecting stocks still below average in a recovering market is more profitable than chasing the recovered winners.

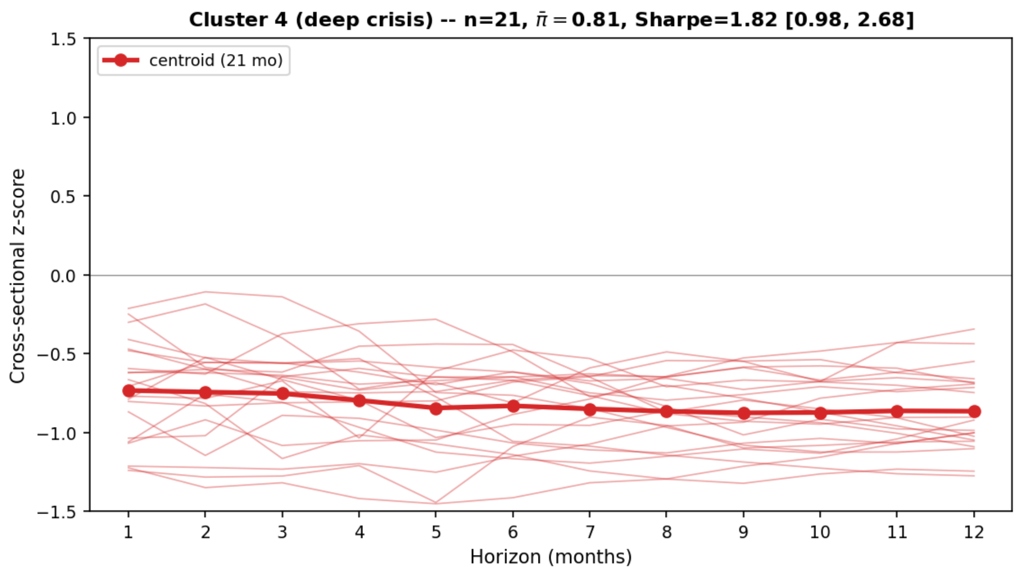


Figure 11: Cluster 4 (deep crisis): z-curves of the 21 member months and centroid. The centroid sits uniformly between −0.73 and −0.88 across all 12 horizons.

Cluster 4 (deep crisis): an active panic with the whole market falling

The long-leg z-curve sits uniformly between −0.73 and −0.88 across all 12 horizons (deepest around month 9, z-mean −0.82), meaning the picks are roughly 0.7 to 0.9 cross-section standard deviations below the average stock at every horizon. Their absolute returns deepen from −15% over the prior month to −48% over the prior 12 months. Eleven of 21 months are in 2022 (Fed-hiking cycle), with corroborating instances in April 2020 (COVID trough), October 2015 and January 2016 (China devaluation, oil collapse), and January 2023.

The market is in active panic, with cluster-mean $\bar{\pi}_t^{\text{filter}} = 0.81$ (the highest of any cluster) and the cross-section uniformly negative (−5%/−11%/−7% over 1/6/12 months). The 6-month return is more negative than the 12-month return, so the older half of the year was roughly flat or positive and the steep decline is concentrated in the most recent months. This signature describes short panics rather than prolonged year-long deteriorations. With π at this extreme, the model commits fully to deep-crisis loser-selection and bets on the same high-β rebound mechanism as cluster 3, fired at the panic peak rather than during recovery, on the premise that the panic resolves quickly; Section 5.3.3 examines what happens when it does not resolve quickly. Cross-section dispersion is the lowest of any cluster at both the 5–8 and 9–12 month horizons (0.35 and 0.45), and 1–4 month skewness is also the lowest (3.67), so stocks are moving down together rather than fanning out.

Cluster 4 earns the highest Sharpe of any cluster (1.82), with a mean monthly return of +3.13%. The rebound bet pays off in 13 of the 21 cluster-4 months (62%), with wins averaging well above losses. The combination of a high hit rate and asymmetric win/loss magnitudes drives the cluster's Sharpe. Cluster 4 is the panic-peak counterpart to cluster 3's post-panic recovery, with the same below-average direction of selection but in a uniformly falling cross-section (avg stock −9%) rather than a recovering one, and at the strongest amplitude in the partition (uniform across all 12 horizons rather than concentrated at months 4 and 12 as in cluster 3). Cluster-level leg-beta and SHAP-share diagnostics for the full $K = 4$ partition are in Appendix H.7.

5.2.3 Discussion

Read together, the four clusters trace a single market cycle. In stages 1 and 2 (continuation), the strategy looks like classical momentum, with above-average winner selection in cluster 1 and a weakened version in cluster 2. In stages 3 and 4 (reversal), it looks nothing like classical momentum, inverting to loser selection in cluster 3 and to a term-structure-wide reversal in cluster 4. The regime-conditioning architecture is therefore not "momentum plus a regime control" but "momentum where momentum works, reversal where momentum crashes". The analysis contributes a regime-conditional mechanism with two interlocking facets to the momentum literature.

XGB ranks stocks on predicted forward return rather than on a per-horizon momentum score, which allows the long leg to hold stocks that are not top-ranked at any individual momentum horizon. This composition is novel for a momentum strategy. Classical (Jegadeesh and Titman, 1993), multi-horizon (Novy-Marx, 2012), and multi-asset (Asness et al., 2013) momentum strategies all construct a momentum score per stock and buy the top decile by that score, so the long-leg names are by construction top-ranked at the scoring horizon. ML-based asset-pricing methods (Gu et al., 2020; Beckmeyer and Wiedemann, 2025) pick top decile by predicted return and could in principle put non-top-momentum names in the long leg, but the published literature does not map picks against the full term structure of past returns. The cluster analysis here shows that XGB does so systematically. In cluster 4 the long leg sits below the cross-section average at every horizon from 1 to 12. In cluster 3 it sits below average at every horizon while the cross-section itself is up. No single-horizon rank-based momentum strategy would select these stocks for the long side.

In calm regimes the model picks above-average z-scores (classical winners). In panic regimes it picks below-average z-scores (reversal), buying stocks uniformly below the cross-section average across all 12 horizons during panic peaks (cluster 4), while keeping the long-short portfolio fully invested. This regime-conditioned direction flip is also novel. Existing crash-management approaches respond to panic by adjusting how much the strategy invests (Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015), how the horizons are blended (Goulding et al., 2023), or by learning regime-agnostic signal-reweightings (Beckmeyer and Wiedemann, 2025). XGB instead reverses which stocks the strategy invests in. Cluster-level CAPM betas confirm the mechanism. XGB's long leg holds high-β stocks in every cluster, including at panic peaks, so when the market rallies the long leg participates in the rebound rather than being dragged down by it (Appendix H.7, Table 27).

5.3 Robustness and Limitations

The strategy's headline performance comes with three limitations. First, the baseline volatility is high (around 19.5%) and some investors will need to scale it down; Section 5.3.1 traces the risk-return trade-off offered by a CRRA risk-aversion control. Second, the 2011–2025 test sample does not include the worst historical stresses; Section 5.3.2 extends the evaluation to 25 years (2000–2024) via expanding-window backtests covering the dot-com bust, the GFC, and the 2009 momentum crash. Third, the panic-regime reversal rests on the premise that price declines during stress are temporary. A sustained economic deterioration would expose the strategy to reversal failure, which Section 5.3.3 discusses alongside the complementary D&M exposure-scaling overlay.

5.3.1 Risk-Return Trade-off

The first limitation is volatility itself. An investor may prefer lower portfolio risk at the cost of some expected return. We incorporate a risk-aversion parameter γ into the strategy and trace the out-of-sample trade-off across $\gamma \in [0, 1]$.

The natural starting point is the mean-variance utility $r - \frac{\gamma}{2}\sigma^2$ (Markowitz, 1952), but it fails here for two reasons. Mechanically, the variance penalty applies to every stock regardless of trading direction, so as γ rises the long leg fills with low-volatility names and the short leg with high-volatility names. The realised long-short volatility then rises rather than falls (Appendix E). Economically, mean-variance penalises absolute volatility, while real investors care about volatility relative to wealth. Constant relative risk aversion captures this proportional treatment, mean-variance does not.

We therefore adopt a CRRA utility scored separately for each trading direction, with $\log \hat{\sigma}$ in place of $\hat{\sigma}^2$: the log penalty makes the cost of risk proportional to its level (fixing the economic issue), and computing utility per-direction makes low-volatility names attractive for either leg (fixing the mechanical issue, so the realised long-short volatility falls as γ rises). For each stock s, the utility under each trading direction is

$$\begin{aligned} u^s_{\text{long}} &= +\,\text{sign}(\hat{r}^s)\,\log\big(1 + |\hat{r}^s|/\varepsilon\big) \;-\; \gamma\,\log(\hat{\sigma}^s), \\ u^s_{\text{short}} &= -\,\text{sign}(\hat{r}^s)\,\log\big(1 + |\hat{r}^s|/\varepsilon\big) \;-\; \gamma\,\log(\hat{\sigma}^s), \end{aligned} \tag{28}$$

where $\hat{r}^s$ is the XGBoost forward-return prediction, $\hat{\sigma}^s$ a forward-volatility forecast from a parallel XGBoost model trained on realised 12-month volatility, and $\varepsilon = 0.01$. The volatility penalty enters with the same sign in both directions, so a volatile stock is equally unattractive whether considered as a holding or as a short position. Each month the model picks stocks that maximise these utilities. Longs are the top decile of u_{long} across the full cross-section, shorts are the top decile of u_{short}. Full specification, including the volatility-forecast model, is in Appendix F.

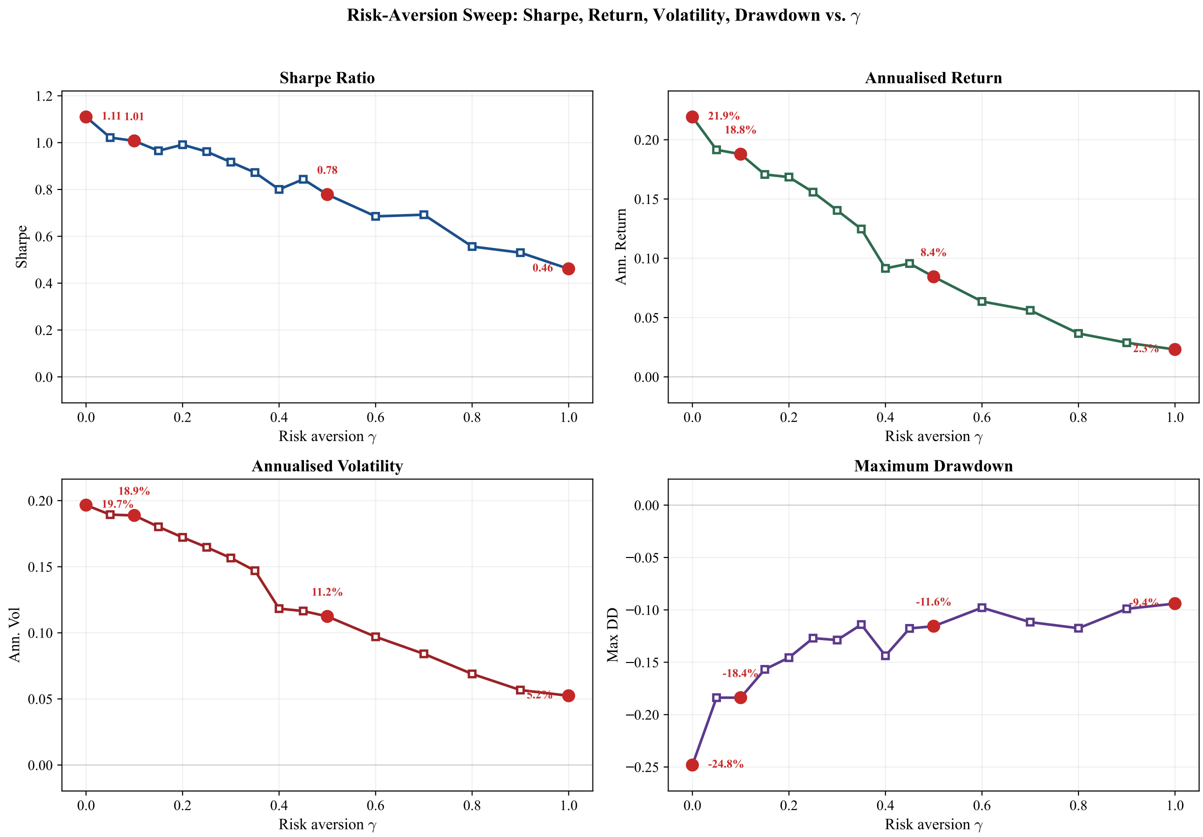


Figure 12: Risk-aversion sweep. Sharpe ratio, annualised return, annualised volatility, and maximum drawdown as a function of γ over $[0, 1]$. Red markers highlight $\gamma \in \{0, 0.1, 0.5, 1.0\}$.

Figure 12 traces the resulting risk-return trade-off as γ rises, where each point corresponds to a different γ and gives a different return-and-risk pair. At $\gamma = 0$ the volatility penalty vanishes and the utilities reduce to a monotonic transformation of the predicted return, so the picks coincide with the headline XGB scoring (Table 3). At $\gamma = 0.1$, Sharpe stays at 1.01 while maximum drawdown tightens from -25% to -18%, a near-free reduction in tail risk. At $\gamma = 0.5$, annualised volatility halves from 19.7% to 11.2%, drawdown halves to -12%, and Sharpe settles at 0.78. At $\gamma = 1.0$, volatility falls to 5.2% (roughly a quarter of the market's) and Sharpe remains positive at 0.46. XGB's regime-conditional edge depends on taking volatility. Its reversal bet, especially cluster 4, loads the long leg with stocks that have fallen the furthest. These are typically the most volatile names in the cross-section, and the bet pays off when they rebound. As γ rises, the volatility penalty filters out exactly these names, and the strategy progressively loses the contrarian bet that generates its panic-regime Sharpe. Beyond $\gamma \approx 1.5$ the long-short spread collapses into dollar-neutral noise, comparable to the zero-Sharpe outcome of the linear baselines on the same features (Section 5.1).

5.3.2 Historical Stress: Out-of-Sample Evidence

The 14-year test sample used so far (2011–2025) misses the worst momentum-relevant stress events: the dot-com bust, the global financial crisis, and the 2009 momentum crash documented by Daniel and Moskowitz (2016). To extend the evaluation, the full pipeline (HMM + XGBoost) is re-estimated on an annual expanding-window schedule: annual retraining is initialised in 1995, with each year's model trained only on data available through 31 December of the prior year and used to make predictions for the following year. Stitching these year-by-year predictions together produces a 25-year out-of-sample monthly return series spanning 2000–2024 (299 months).[22] Methodology details are in Appendix H.2.

		XGB			Market		
Period	N months	Sharpe	Total Return	Max DD	Sharpe	Total Return	Max DD
Full sample, 2000–2024	299	0.50	898.4%	−64.8%	0.56	572.6%	−50.3%
2000–2002 dot-com	36	−0.15	−35.9%	−64.8%	−0.72	−38.1%	−45.0%
2003–2006 bull	48	0.70	37.5%	−12.7%	1.70	80.5%	−4.6%
2007–2009 GFC	36	0.26	11.1%	−36.3%	−0.15	−13.9%	−50.3%
2010–2019 calm	120	0.67	155.4%	−17.9%	1.05	258.1%	−17.6%
2020–2024 COVID era	59	1.23	299.1%	−22.2%	0.81	95.2%	−24.8%

Table 9: Sub-period decomposition of the expanding-window out-of-sample backtest, 2000–2024. Annual retraining is initialised in 1995 using all data available through the prior December. Market is the value-weighted CRSP market (Mkt-RF + RF, Fama-French monthly factors).

Worst observed stress: the dot-com bust

The 2000–2002 dot-com bust produced the strategy's largest historical drawdown of −64.8% peak-to-trough. Restricted to the bear-market proper (March 2000–October 2002), the strategy lost −43.8% cumulative over 32 months (Sharpe −0.39). This is the historical realisation of the reversal-failure vulnerability discussed in Section 5.3.3 below. A prolonged sector-rotating bear in which beaten-down stocks did not promptly recover, so the panic-regime reversal selection bled.

Stress at the targeted failure mode: the 2009 momentum crash

The single most informative episode is March–December 2009, the period Daniel and Moskowitz (2016) document as a severe momentum crash. Under the same long-short construction used throughout this thesis (NYSE breakpoints, value-weighted, 10 bps cost), unconditional 12-month momentum lost −62.6% cumulative during those ten months. Under expanding-window out-of-sample, the regime-conditional strategy gained +20.5% cumulative (Sharpe 0.73) during the same window. Across the full GFC sub-window

[22] The expanding-window construction enforces strict out-of-sample status throughout: each year's predictions are produced by a model trained only on observations available at the corresponding cutoff, so no future information leaks into any prior-year prediction.

(2007–2009 in Table 9) the strategy returned +11.1% (Sharpe 0.26) while the market lost −13.9% (Sharpe −0.15). This is direct historical evidence that the regime mechanism survives the specific stress event that defeats both unconditional momentum and the broader market.

5.3.3 Reversal Failure and the D&M Complement

The dot-com episode above is the empirical realisation of the strategy's reversal-failure mode. The cluster 4 bet rests on the premise that panic-period declines are temporary, which holds in shorter panics; XGB earned +29.9% during the COVID crash (February–April 2020) and +21.5% during the 2022 bear market (January–October 2022), and in both cases the decline reversed within months. The premise fails in a sustained economic deterioration where beaten-down stocks do not promptly recover because their businesses are failing, as in the 32-month dot-com bear (March 2000–October 2002) where XGB lost −43.8% cumulative with a peak-to-trough drawdown of −64.8%. The regime classification was correct. The within-regime momentum structure had changed.

XGB and Daniel and Moskowitz (2016) use regime information along complementary channels. XGB uses it to choose *which* stocks to hold; D&M use it to scale *how much* to hold. On the same data and transaction costs, XGB (Sharpe 1.11, CAPM alpha 23.4%) outperforms D&M managed momentum (Sharpe 0.49, CAPM alpha 10.3%).[23] The two approaches are not substitutes. A hybrid that adds a D&M-style exposure-scaling overlay on top of XGB as a circuit breaker (Section 6.2) would reduce the risk of a total collapse during a prolonged bear market like the dot-com bust without sacrificing XGB's stock-selection advantage in shorter panics.

5.3.4 International Robustness

To test whether the cross-sectional regime-momentum mechanism extends beyond US data, we replicate the full pipeline on UK and Japanese equities (Compustat Global, 1990–2025).[24]

[23] The 0.49 figure is the D&M dynamic-weighting strategy reimplemented on this thesis's 2011–2025 U.S. equity sample with the same transaction cost assumptions as XGB, not the 1.19 figure reported in Daniel and Moskowitz (2016) (Table 7) for the 1934–2013 sample. The lower Sharpe on the post-2010 sample is consistent with the broader weakening of unconditional momentum since 2009, not a reimplementation discrepancy.

[24] Each region's HMM is fit on locally-selected stress features (DD+VOL+REL_N for both, with BANK_REL replacing the US-specific Moody's BAA-AAA spread in the candidate pool). Details in Appendix K. The cross-sectional model is unchanged from the US specification, except that long and short legs are formed from unconditional decile breakpoints across all listed stocks rather than NYSE-equivalent breakpoints (no analogous tier exists for either market).

Strategy	UK				JP			
	Sharpe	Ann. Ret	Total Ret	Max DD	Sharpe	Ann. Ret	Total Ret	Max DD
Local market	0.62	6.7%	170.6%	−25.2%	0.86	12.3%	493.2%	−21.5%
Fixed 12-mo mom	0.45	9.3%	289.9%	−62.6%	0.07	−0.5%	−6.9%	−66.3%
XGB	0.84	16.0%	864.6%	−21.4%	0.50	6.0%	143.2%	−19.8%

Table 10: International cross-sectional model performance, 2011–2025 OOS. Regionally-fitted HMM (DD+VOL+REL_N).

Both regions support the mechanism. UK XGB attains a Sharpe of 0.84 over 2011–2025, well above unconditional 12-month momentum's 0.45. JP XGB attains 0.50, substantially above unconditional 12-month momentum's 0.07, consistent with the documented weakness of Japanese momentum (Asness et al., 2013). Maximum drawdown is reduced from −62.6% (fixed momentum) to −21.4% in UK, and from −66.3% to −19.8% in JP.

The mechanism therefore transfers across markets, but with regional variation. In the UK, the regime-conditional model dominates unconditional momentum on every dimension. The mechanism works in the same form as in the US. In Japan, XGB clearly improves on unconditional momentum (Sharpe rises from 0.07 to 0.50 and total return from −6.9% to +143.2%). Read together with the US result, this pattern indicates that the regime-conditional cross-sectional mechanism is real and portable.

6 Summary, Conclusions and Recommendations

Summary

This thesis examined the behaviour of cross-sectional momentum across market regimes. While the existing literature establishes that momentum is regime-dependent, this thesis maps out how that dependence is shaped. The empirical strategy paired a Bayesian Hidden Markov regime filter, built from monthly stress indicators, with cross-sectional stock-selection models of increasing complexity that operate on the regime signal and the full momentum term structure. The framework was evaluated out-of-sample on U.S. equities (2011–2025), opened up with SHAP and cluster analysis to characterise the role the regime signal plays in selection, and tested outside the main sample on a 25-year expanding-window backtest covering the dot-com bust, the GFC, and the 2009 momentum crash, and on UK and Japanese equities with locally-fitted HMMs. The remainder of this chapter draws conclusions from these results, discusses their limitations, and recommends directions for future research.

Conclusions and Contributions

The mapping yields a clear picture. Cross-sectional momentum's regime-dependent structure is real and characterisable, and the mechanism is that the regime signal acts as a context variable. It has near-zero direct return-predictive content, yet it conditions how a nonlinear cross-sectional model uses the full momentum term structure in a given market state. The empirical evidence is consistent with this account. A nonlinear model produces positive out-of-sample alpha while the linear baseline on identical features collapses, the regime signal is essential to the result, and the mechanism replicates internationally and survives historical stress events outside the main test sample (Section 5).

The thesis makes two contributions to the momentum literature. First, the methodological combination is itself novel. Prior regime-conditional momentum work has paired simple regime treatments (binary up-or-down indicators, single-variable volatility thresholds, or fast-slow blending weights) with linear or rank-based cross-sectional rules. This thesis instead pairs a continuous probabilistic regime filter built from multiple stress indicators (the Bayesian HMM filtered probability π_t^{filter}) with a nonlinear cross-sectional model (XGBoost), and the combination is what enables the empirical mechanism. The filtered probability acts as a routing variable through tree splits, conditioning the joint use of all twelve momentum horizons on the current market state in a way that no linear or single-horizon rule can represent.

Second, the analysis of how that mechanism operates reveals two interlocking facets of a regime-conditional mechanism (Section 5.2.3). XGB ranks stocks on predicted forward return rather than per-horizon momentum rank, which allows the long leg to hold stocks not top-ranked at any individual horizon. This is a composition that no single-horizon rank-based momentum strategy can produce by construction, observed in clusters 3 and 4 where the long leg sits below the cross-section average at every horizon (cluster 3 with the cross-section itself rallying, cluster 4 at the panic peak). In calm regimes the model picks above-average z-scores. In panic regimes it picks below-average z-scores, while keeping the long-short portfolio fully invested. Cluster-level CAPM betas confirm the mechanism: XGB's long leg holds high-β stocks in every cluster, including at panic peaks, so when the market rallies the long leg participates in the rebound rather than being dragged down by it (Appendix H.5). The three approaches in the existing literature (simple-rank momentum selection, exposure scaling, and selection on a learned reweighting of past returns) are complementary to this one, and a hybrid combining XGB's stock-selection with a D&M-style exposure-scaling overlay would address the reversal-failure vulnerability without sacrificing XGB's stock-selection advantage in shorter panics.

6.1 Limitations

Reversal-failure vulnerability

The strategy's primary vulnerability is a prolonged economic deterioration in which the prior decline reflects genuine business failure rather than temporary mispricing, so the beaten-down stocks do not recover. The dot-com bust is its historical realisation (Section 5.3.2). A sustained multi-year decline without any reversal (a depression-style outcome) lies outside the 25-year historical sample.

Test period scope

The 2011–2025 test period does not contain a prolonged recession. Performance is strongest in 2021–2025, a period with frequent regime transitions. In calm-dominated years the model earns modest returns from standard continuation. A permanently calm market would reduce the strategy to ordinary momentum.

Frozen HMM parameters and regime drift

The HMM is estimated on 1990–2010 and held fixed during the test period. This assumes that the statistical properties of calm and panic regimes remain stationary. If the market undergoes structural change (a permanent increase in baseline volatility due to algorithmic trading or shifting monetary policy regimes, for example), the trained thresholds may become miscalibrated, with structurally calm periods classified as panic. The sustained elevation of π_t^{filter} in recent years may partly reflect such drift rather than genuine prolonged stress. The 25-year expanding-window backtest in Section 5.3.2 (in which the HMM is re-estimated annually using only data available at the time) reproduces the production result on the 2011–2024 overlap, indicating that the regime structure is reasonably stable. Online re-estimation would still provide a stronger guarantee.

Specification search

The HMM feature selection procedure uses out-of-sample portfolio Sharpe in its second pass, meaning the selected combination (DD+DISP+REL_N+CS) is informed by test-period outcomes. Section 4.2 discusses the mitigating factors in full (the quality screen in Pass 1, the cross-seed stability criterion in Pass 4, and the qualitatively similar performance of leave-one-out specifications, Table 13). But the reported 1.11 Sharpe should be read with this caveat.

6.2 Recommendations for Future Research

Four directions extend this framework most directly.

Richer regime-signal representation

The current model uses only the contemporaneous filtered probability π_t^{filter}, discarding regime dynamics. Two complementary extensions follow. The first looks forward, computing a k-step regime forecast $P(s_{t+k} = 1 \mid \mathbf{z}_{1:t})$ from the HMM's transition probabilities so the portfolio can reposition before a regime shift rather than after it. The gain is concentrated in transition months, where the current framework trades on a stale view of the regime. The second looks backward, feeding the trajectory of π_t^{filter} rather than its current value so that a fresh spike (where mean reversion is most likely) and a sustained elevation (where the reversal mechanism may be breaking down) become distinguishable. A sequential model such as an LSTM or transformer, or simple lag-based feature engineering (lagged π_t^{filter} values, a short-window rate of change, a regime-duration counter), would expose this trajectory. Together, the two extensions provide anticipation and memory.

Alternative nonlinear models

XGBoost is one of several nonlinear architectures, and the central mechanism (that π_t^{filter} gates the use of momentum horizons) should in principle be recoverable by any sufficiently expressive model. Gu et al. (2020) benchmark a broad set of machine learning methods for cross-sectional return prediction and find tree ensembles and neural networks of moderate depth (peaking at three hidden layers) most competitive, making a feedforward network the natural next architecture to test alongside the XGBoost ensemble used here. A feedforward network with π_t^{filter} and the twelve momentum horizons as inputs is the simplest baseline. An attention-based model would learn horizon-by-regime weights directly and produce an interpretable per-horizon attention profile comparable against the SHAP decomposition in Table 7. A recurrent or transformer-based variant would naturally absorb the trajectory extension proposed above. Recovering the mechanism across these architectures would strengthen the claim that the regime-conditional shift is a property of the data rather than of XGBoost specifically.

Regime-conditional feature sets

Adding ten firm-level fundamentals to XGB reduces Sharpe (Section 5.1.3) because the new features absorb a substantial share of SHAP importance and dilute the regime-momentum signal. The fund-only variant matches the baseline's calm performance but produces a far weaker panic Sharpe, because firm-level characteristics describe stable attributes that do not flip direction when the regime shifts. The optimal feature set may itself be regime-dependent. During panic, when reversal dominates, momentum and the regime signal carry the mechanism. During calm, when the continuation premium is weaker and more diffuse, fundamentals may provide a useful second dimension. A layered construction in which XGB produces the initial long-short ranking and fundamentals enter as a post-ranking filter (excluding bottom-quintile-quality names from the long leg, high-leverage names from the short leg) is a natural starting point. A more ambitious variant trains separate cross-sectional models for the calm and panic branches, each with its own feature set. Both designs require careful out-of-sample validation because the regime-conditional split reduces effective sample size.

Exposure-scaling overlay (D&M complement)

An exposure-scaling overlay layered on top of XGB's stock-selection requires a triggering rule. Natural candidates are π_t^{filter} above a threshold for a minimum number of consecutive months, or its rolling maximum over a fixed window exceeding a calibrated bound. Preliminary checks on the 2011–2025 sample show no improvement from the overlay because the sample contains no prolonged recession, so the circuit breaker only cuts profitable panic months. Its value would emerge in a sustained downturn outside the current window. The dot-com sub-period of the 25-year expanding-window backtest (Section 5.3.2) is a natural setting to evaluate candidate overlay rules.

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Appendix A: Data Sources and Variable Definitions

Three primary data sources are used. **CRSP** (Center for Research in Security Prices, via WRDS) provides monthly stock-level returns, prices, shares outstanding, exchange codes, and share codes from the Monthly Stock File (`crsp.msf`), as well as daily market returns from the Daily Stock Index file (`crsp.dsi`). **Compustat** (via WRDS) provides quarterly fundamental accounting data from `comp.fundq`, linked to CRSP via the CCM bridge table using primary link types (LU, LC). **FRED** (Federal Reserve Economic Data) provides Moody's BAA and AAA corporate bond yields for computing the credit spread.

Following Shumway (1997), delisting returns are incorporated to prevent survivorship bias. When a stock is delisted, its actual delisting return is used if available; for performance-related delistings (CRSP codes 500–584) with missing returns, −30% is imputed.

A comprehensive overview of all data variables is provided in Table 11.

Variable	Source	Description	Used in
Stock-level variables			
Monthly returns	CRSP `msf`	Adjusted holding-period returns	Sections 3, 5
Price, shares out.	CRSP `msf`	For market equity computation	Section 4
Exchange, share codes	CRSP `msf`	Universe filtering (codes 10, 11)	Section 3.2
Delisting returns	CRSP `msedelist`	Survivorship bias correction	Section 3.2
Book equity (`ceqq`)	Compustat `fundq`	Book-to-market, ROE	Appendix L
Total assets (`atq`)	Compustat `fundq`	Asset growth, gross profit, CF ratios	Appendix L
Income (`ibq`)	Compustat `fundq`	ROE, earnings growth, accruals	Appendix L
Revenue (`revtq`)	Compustat `fundq`	Gross profitability	Appendix L
COGS (`cogsq`)	Compustat `fundq`	Gross profitability	Appendix L
Debt (`dlttq`, `dlcq`)	Compustat `fundq`	Leverage ratio	Appendix L
Operating cash flow (`oancfy`)	Compustat `fundq`	CFO, FCF, accruals	Appendix L
Capital expenditure (`capxy`)	Compustat `fundq`	Free cash flow	Appendix L
Market-level variables (HMM inputs)			
Daily market returns	CRSP `dsi`	Drawdown (DD) computation	Section 4.1
Market price index	CRSP `dsi`	Drawdown (DD)	Section 4.1
Cross-sect. stock returns	CRSP `msf`	Dispersion (DISP), REL_N	Section 4.1
BAA, AAA bond yields	FRED	Credit spread (CS)	Section 4.1
Derived features			
$mom_1, \ldots, mom_{12}$	Computed	12 trailing momentum signals	Section 4.4
π_t^{filter}	HMM output	Filtered panic probability	Section 4.3
10 fundamentals (ablation only)	Computed	bm, roe, gross profit, cash flow, etc.	Appendix L

Table 11: Summary of all data variables used in the analysis.

Notes: All data accessed via WRDS. Compustat linked to CRSP via the CCM bridge table using primary link types (LU, LC). Fundamentals are lagged four months from fiscal quarter end to ensure real-time availability. The sample covers January 1990 to November 2025 (training: 1990–2010; test: 2011–2025).

Appendix B: HMM Feature Selection Tables

The two tables in this section support the four-pass feature selection procedure described in Section 4.2. Table 12 reports the top 10 candidate feature combinations on ensemble Sharpe and cross-seed stability. Table 13 reports include-one and leave-one-out ablations of the production four-feature set.

D	Combination	Sharpe	Mean	σ
4	DD+DISP+REL_N+CS	0.91	0.84	0.11
4	DD+DISP+CS+TERM	0.82	0.55	0.15
4	DD+DISP+REL_N+SKEW	0.80	0.69	0.08
4	DD+VOL+DISP+REL_N	0.80	0.77	0.09
3	DD+DISP+REL_N	0.77	0.77	0.10
4	DD+VOL+DISP+TERM	0.71	0.67	0.16
3	DD+VOL+CS	0.69	0.43	0.13
1	DD	0.68	0.67	0.14
4	DD+DISP+REL_N+TERM	0.62	0.45	0.18
3	DD+ADR+TERM	0.57	0.69	0.07

Table 12: Top 10 feature combinations from the three-pass selection pipeline (long-short, 10 bps costs). D = number of HMM features. Sharpe = ensemble Sharpe ratio (10 HMM seeds, 20 XGBoost seeds). Mean and σ are the mean and standard deviation of the Sharpe ratio across 5 independent experiments with different seed partitions, measuring robustness to random initialisation. DD+DISP+REL_N+CS ranks first on both ensemble Sharpe and stability mean.

HMM Input Features	XGB Sharpe	Δ vs Full
Full 4F baseline (DD+DISP+REL_N+CS)	0.97	—
Include-one (single-feature HMM)		
DD only (market drawdown)	0.66	−0.31
Leave-one-out (three-feature HMM)		
Drop DD	0.51	−0.46
Drop DISP	0.96	−0.01
Drop REL_N	0.43	−0.54
Drop CS	0.96	−0.01

Table 13: HMM input feature ablation (long-short, feature-selection seed counts). The absolute Sharpe levels differ from the production model (Table 3) due to fewer seeds; the relative comparisons (Δ) are the relevant quantities. DD alone nearly matches the full model. Removing DD or REL_N causes the largest drops (−0.46 and −0.54), while DISP and CS contribute marginally but improve seed stability (Section 4.2).

Appendix C: Portfolio Formation, Transaction Costs, and Performance Metrics

C.1 Portfolio Construction

For each method, a long-short portfolio is constructed each month. Stocks are ranked by their scores, and NYSE-listed stocks are used to determine decile breakpoints. All stocks (including AMEX and NASDAQ) whose score exceeds the 90th percentile breakpoint enter the long leg; all stocks below the 10th percentile breakpoint enter the short leg. Long-short construction is the standard in the momentum literature (Jegadeesh and Titman, 1993; Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015) because it isolates momentum's cross-sectional stock selection from market beta, providing a cleaner test of the strategy's ability to distinguish future winners from losers.

Portfolio weights within each leg are assigned proportionally to market equity (value-weighting):

$$w_t^{s,L} = \frac{ME_t^s}{\sum_{s\in\mathcal{L}_t} ME_t^s}, \qquad w_t^{s,S} = \frac{ME_t^s}{\sum_{s\in\mathcal{S}_t} ME_t^s} \tag{29}$$

where $\mathcal{L}_t$ and $\mathcal{S}_t$ are the sets of stocks in the long and short legs at month t, respectively. The long-short portfolio return is:

$$r_t^{\mathrm{LS}} = \sum_{s\in\mathcal{L}_t} w_t^{s,L} \cdot r_t^s - \sum_{s\in\mathcal{S}_t} w_t^{s,S} \cdot r_t^s \tag{30}$$

Value-weighting tilts each leg toward large, liquid stocks, reducing the influence of small-cap return anomalies and improving implementability.

C.2 Transaction Costs

All portfolios are rebalanced monthly.

A one-way transaction cost of $c = 10$ basis points is applied to the turnover in both the long and short legs at each rebalancing date. The net portfolio return at month t is:

$$r_t^{\mathrm{net}} = r_t^{\mathrm{gross}} - c \cdot (\tau_t^L + \tau_t^S) \tag{31}$$

where τ_t^L and τ_t^S are the one-way turnover in the long and short legs, respectively:

$$\tau_t^L = \frac{1}{2}\sum_s \left| w_t^{s,L,\mathrm{new}} - w_{t-1}^{s,L,\mathrm{prev}} \right|, \qquad \tau_t^S = \frac{1}{2}\sum_s \left| w_t^{s,S,\mathrm{new}} - w_{t-1}^{s,S,\mathrm{prev}} \right| \tag{32}$$

and each summation runs over all stocks appearing in either the new or previous portfolio for that leg. The 10 basis point cost reflects the transaction environment for a value-weighted, large-cap-biased strategy where NYSE breakpoints tilt toward liquid securities.

It is important to note that transaction costs are applied *post-hoc* as an accounting adjustment and are not incorporated into the models' training objectives. The cross-sectional models (DET, LR, and XGB) optimise stock-level prediction (classification accuracy or return forecasting) without knowledge of the previous portfolio, the resulting turnover, or the cost of rebalancing. This is standard practice in the empirical asset pricing literature (e.g., Fama and French, 2015; Goulding et al., 2023), where portfolio construction is treated as a fixed downstream step applied uniformly to all strategies. An end-to-end framework that jointly optimises stock selection and portfolio-level objectives (e.g., net-of-cost Sharpe ratio) would require a fundamentally different architecture, such as a reinforcement learning or differentiable portfolio optimisation layer, and remains a potential extension of this work.

C.3 Performance Metrics

Standard Metrics

Strategy performance is assessed using four standard risk-adjusted metrics computed on monthly net-of-fee returns.

Annualized return The geometric annualized return over n months is:

$$R_{\text{ann}} = \left[\prod_{t=1}^{n}(1 + r_t)\right]^{12/n} - 1 \tag{33}$$

Annualized volatility The annualized standard deviation of monthly returns:

$$\sigma_{\text{ann}} = \sigma(r) \cdot \sqrt{12} \tag{34}$$

Sharpe ratio The annualized Sharpe ratio:[25]

$$SR = \frac{\bar{r}}{\sigma(r)} \cdot \sqrt{12} \tag{35}$$

where $\bar{r}$ is the mean monthly return. Sharpe is computed from the arithmetic monthly mean annualised by $\sqrt{12}$, while annualised returns reported elsewhere in this thesis are compound (geometric); under high volatility the two definitions of average return differ, so Sharpe should not be inferred from a table by dividing annualised return by annualised volatility (e.g. at depth 1 in the methodology hyperparameter sweep, the reported Sharpe

[25] All Sharpe ratios in this thesis are computed using raw returns rather than excess returns above the risk-free rate. Since the same convention is applied to every strategy, relative comparisons and rankings are unaffected. The risk-free rate was near zero for most of the test period (2011–2021) and averaged roughly 2% annualized over the full 2011–2025 window, so absolute Sharpe ratios are overstated by approximately 0.05–0.10.

is 0.53 but 8.7%/19.4% = 0.45).

Maximum drawdown Let $C_t = \prod_{\tau=1}^{t}(1 + r_\tau)$ denote the cumulative wealth path. The maximum drawdown is:

$$MDD = \min_{1\leq t\leq n} \frac{C_t - \max_{1\leq\tau\leq t} C_\tau}{\max_{1\leq\tau\leq t} C_\tau} \tag{36}$$

Regime-Conditional Evaluation

To assess whether the regime signal adds value beyond unconditional stock selection, performance is decomposed by market regime. Months are partitioned into calm ($\pi_t^{\text{filter}} < 0.5$) and panic ($\pi_t^{\text{filter}} \geq 0.5$) periods, and the Sharpe ratio is computed separately within each subset:

$$SR_{\text{calm}} = \frac{\bar{r}_{\text{calm}}}{\sigma(r_{\text{calm}})} \cdot \sqrt{12}, \qquad SR_{\text{panic}} = \frac{\bar{r}_{\text{panic}}}{\sigma(r_{\text{panic}})} \cdot \sqrt{12} \tag{37}$$

This decomposition helps assess whether the strategy's alpha originates primarily from calm-period stock selection, panic-period risk avoidance, or both.

Appendix D: Feature Importance Methods

D.1 SHAP Values for XGBoost

To interpret the XGBoost model and assess the marginal contribution of each feature, SHAP (SHapley Additive exPlanations; Lundberg and Lee, 2017) values are computed on the test set. The SHAP value of feature j for observation $\mathbf{x}$ measures its average marginal contribution to the prediction across all possible feature orderings:

$$\phi_j(\mathbf{x}) = \sum_{S\subseteq N\setminus\{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \Big[f_{S\cup\{j\}}(\mathbf{x}) - f_S(\mathbf{x})\Big] \tag{38}$$

where N is the full feature set, $p = |N|$, and $f_S(\mathbf{x})$ is the expected model output when only features in S are observed. For tree-based models, the TreeSHAP algorithm (Lundberg et al., 2020) computes these values exactly in polynomial time. The overall importance of each feature is summarised as:

$$I_j = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} |\phi_j(\mathbf{x}_i)| \tag{39}$$

Two additional analyses are performed. First, the mean absolute SHAP value of π_t^{filter} is computed separately for calm and panic months to assess whether the regime signal's influence varies across market states. Second, regime-specific XGBoost models are trained on calm-only and panic-only subsets of the training data, and their SHAP feature importances are compared to identify which features the model relies on differentially across regimes.

Appendix E: Alternative Training Targets for XGBoost

Section 5.3 argues that embedding a risk penalty directly in the XGBoost training target degrades performance, motivating the CRRA specification used in the main text. Table 14 shows this degradation is not an artefact of the mean-variance functional form: the same pattern appears under three families of training targets with varying risk tolerance.

Target family	Parameter	Sharpe	Ann. Ret	Ann. Vol	MDD
Baseline (r)	–	1.107	30.0%	27.1%	−28.3%
Mean-variance: $r - \frac{\gamma}{2}\sigma^2$	$\gamma = 0.5$	1.061	22.9%	21.6%	−22.3%
	$\gamma = 1$	1.024	16.5%	16.2%	−22.8%
	$\gamma = 2$	1.135	15.5%	13.6%	−22.0%
	$\gamma = 5$	1.062	13.5%	12.7%	−22.5%
	$\gamma = 10$	1.082	13.1%	12.1%	−18.9%
Log mean-variance: $\log(1+r) - \frac{\gamma}{2}\sigma^2$	$\gamma = 0.5$	1.051	14.6%	13.9%	−17.9%
	$\gamma = 1$	1.039	13.8%	13.4%	−23.8%
	$\gamma = 2$	1.080	14.0%	12.9%	−20.9%
	$\gamma = 5$	1.127	14.1%	12.4%	−18.7%
	$\gamma = 10$	0.995	12.1%	12.3%	−20.2%
Sharpe-like: $r\,/\,\sigma^a$	$a = 0.5$	1.148	24.8%	21.4%	−22.4%
	$a = 1.0$	0.984	14.6%	15.0%	−29.1%
	$a = 1.5$	0.936	12.4%	13.5%	−21.0%
	$a = 2.0$	0.759	12.8%	18.1%	−24.5%

Table 14: Out-of-sample performance of XGBoost (XGB) under alternative training targets (long-only construction). Every risk-adjusted target reduces returns faster than volatility.

The pattern is consistent: returns fall faster than volatility under every risk penalty, regardless of functional form. The mildest adjustments (mean-variance with $\gamma = 0.5$, Sharpe-like with $a = 0.5$) come closest to the baseline but still underperform. This confirms that the diagnostic in Section 5.3 is not specific to the mean-variance utility formulation; the single-model training-target approach is structurally unsuited to incorporating risk aversion in a long-short setting, which the CRRA dual-utility framework resolves.

Appendix F: CRRA Dual-Utility Construction

This appendix provides the technical specification behind the risk-aversion sweep in Section 5.3. The strategy uses two forecasts for each stock-month: an expected-return forecast $\hat{r}_t^s$ and a forward-volatility forecast $\hat{\sigma}_t^s$. $\hat{r}_t^s$ is produced by the XGBoost ensemble described in Section 3. $\hat{\sigma}_t^s$ is produced by a second XGBoost ensemble trained on the realised forward volatility

$$\sigma^s_{[t+1,\,t+12]} \;=\; \mathrm{std}\big(r^s_{t+1}, \ldots, r^s_{t+12}\big)$$

using the same feature set augmented with trailing 12-month realised volatility.

For each stock, utility is computed under both trading directions:

$$\begin{aligned} u^s_{\text{long}} &= +\,\mathrm{sign}(\hat{r}^s)\,\log\big(1 + |\hat{r}^s|/\varepsilon\big) \;-\; \gamma\,\log(\hat{\sigma}^s), \\ u^s_{\text{short}} &= -\,\mathrm{sign}(\hat{r}^s)\,\log\big(1 + |\hat{r}^s|/\varepsilon\big) \;-\; \gamma\,\log(\hat{\sigma}^s), \end{aligned} \tag{40}$$

with $\varepsilon = 0.01$. The return term is $\pm$ log-compressed expected return under each direction; the $-\gamma\log(\hat{\sigma})$ term enters with the same sign in both, so a low-σ stock is preferred whether considered as a long or as a short.

Each month, longs are the top decile of u_{long} and shorts are the top decile of u_{short}, both using NYSE breakpoints and full-universe membership. Positions are value-weighted within each leg. Stocks appearing in both top deciles (the overlap set, which grows with γ) are retained on both sides; their contributions to the long and short books net in $r_L - r_S$. This is the mechanism by which the long-short spread converges to zero as γ rises beyond the useful range: at high γ, the volatility penalty dominates the return signal, low-$\hat{\sigma}$ stocks occupy both top deciles simultaneously, and the two legs become near-identical value-weighted low-vol portfolios.

The 50-seed ensemble is trained on 1990–2010 and evaluated out-of-sample on 2011–2025. The sweep in Figure 12 uses a 17-point grid over $[0, 1]$ (step 0.05 up to 0.5, step 0.1 thereafter).

Appendix G: Robustness Checks

This section reports six robustness checks that complement the main results in Section 5.

G.1 Sub-Period Stability

To assess whether the main findings are driven by a single favourable sub-period, the test sample is split into three roughly equal windows: 2011–2015, 2016–2020, and 2021–2025.

	2011–2015	2016–2020	2021–2025	Full
Market	0.72	1.04	0.73	0.84
Fixed 12-mo	0.74	-0.20	-0.11	0.06
LR	0.47	-0.02	-0.38	-0.01
XGB	0.59	1.04	1.70	1.11

Table 15: Sub-period Sharpe ratios (long-short). The test period is split into three roughly equal sub-periods. XGB is the only strategy with positive Sharpe ratios in all three sub-periods.

In long-short construction, XGB is the only strategy that produces positive Sharpe ratios in every sub-period (0.59, 1.04, 1.70). LR collapses in 2016–2020 (Sharpe −0.02) and turns negative in 2021–2025 (−0.38). Fixed 12-month momentum turns negative after 2015, consistent with the momentum crash mechanism documented by Daniel and Moskowitz (2016). XGB's strongest performance (1.70) is in 2021–2025, a period characterised by elevated volatility and multiple stress episodes, precisely the conditions where regime-conditional stock selection adds the most value.

G.2 Turnover and Transaction Cost Sensitivity

A strategy's net-of-cost performance depends critically on its turnover. Table 16 reports average monthly one-way turnover for each strategy.

Strategy	Avg. Monthly TO	Annualised TO
Fixed 12-mo	70.1%	842%
Fixed 1-mo	184.3%	2,211%
LR	168.8%	2,026%
XGB	132.3%	1,588%

Table 16: Portfolio turnover by strategy (long-short). Average monthly one-way turnover and annualised turnover (monthly × 12).

In long-short construction, XGB's average monthly one-way turnover is 132.3%, reflecting the regime-conditional changes in stock selection that shift the portfolio composition substantially. Table 17 translates these turnover differences into Sharpe ratio sensitivity across a range of transaction cost assumptions.

	0 bps	5 bps	10 bps	20 bps	30 bps	50 bps
Fixed 12-mo	0.10	0.08	0.07	0.03	0.00	-0.06
Fixed 1-mo	0.39	0.32	0.26	0.14	0.01	-0.24
LR	0.12	0.05	-0.02	-0.16	-0.29	-0.57
XGB	1.19	1.15	1.11	1.03	0.95	0.78

Table 17: XGB remains profitable at 50 bps. All other strategies become unprofitable at moderate cost levels.

XGB remains profitable across all cost levels tested: even at 50 bps one-way (five times the baseline), its Sharpe ratio is 0.78. All other strategies become unprofitable at moderate cost levels. The baseline 10 bps assumption is reasonable for a value-weighted strategy using NYSE breakpoints, which tilts toward the most liquid names.

G.3 Model Sensitivity

The XGBoost model uses hyperparameters selected via cross-validation on the training set. To verify that the results are not an artefact of a particular configuration, Table 18 reports out-of-sample performance under alternative tree depths, learning rates, and ensemble sizes.

Configuration	Sharpe	Ann. Ret	Ann. Vol
depth=3, lr=0.05, n=500	0.98	19.2%	19.9%
depth=4, lr=0.05, n=500	1.11	21.7%	19.5%
depth=5, lr=0.05, n=500	0.92	17.4%	19.4%
depth=6, lr=0.05, n=500	0.90	17.0%	19.5%
depth=4, lr=0.01, n=500	0.99	18.7%	19.2%
depth=4, lr=0.10, n=500	1.00	19.6%	19.9%
depth=4, lr=0.05, n=200	1.06	20.5%	19.4%
depth=4, lr=0.05, n=1000	0.96	18.7%	19.9%

Table 18: XGBoost hyperparameter sensitivity (long-short). Each row varies one hyperparameter from the baseline. The baseline is not the single best configuration, but all variants outperform unconditional momentum.

Hyperparameter sensitivity

At the baseline depth of 4, the Sharpe ratio ranges from 0.89 to 1.11 across learning rate and ensemble size variations. Performance degrades at depths below 3 (see Section 3.2.1 for the depth analysis) and above 5 due to overfitting. The results confirm that XGB's performance is not an artefact of a particular hyperparameter configuration.

G.4 Number of Hidden States

The baseline HMM uses $K = 2$ states (calm and panic). A natural question is whether finer regime granularity (for example, distinguishing mild corrections from severe crises) would improve the downstream portfolio models. Table 19 reports out-of-sample Sharpe ratios for $K = 2, 3, 4, 5$, each averaged over five representative Gibbs sampler seeds (the production regime signal averages 200 seeds, but five suffice for this diagnostic).

	LR		XGB	
K	Mean Sharpe	Std	Mean Sharpe	Std
2	-0.012	0.000	1.107	0.000
3	-0.029	0.004	0.831	0.164
4	-0.029	0.003	0.606	0.291
5	-0.028	0.003	0.724	0.087

Table 19: Out-of-sample Sharpe ratios for HMMs with $K = 2, 3, 4, 5$ states (long-short). Each entry reports the mean and standard deviation across 5 independent Gibbs sampler seeds (2,000 iterations each). LR is invariant to K. XGB shows no consistent improvement beyond $K = 2$, and cross-seed variability increases with K.

In long-short construction, XGB shows no consistent improvement beyond $K = 2$. Cross-seed variability increases with K, and the intermediate states that emerge for $K \geq 3$ lack the persistence needed to provide a stable conditioning signal. With only 241 training months, the data cannot reliably separate more than two regimes for the purpose of downstream cross-sectional stock selection.

Beyond estimation, $K = 2$ has an interpretability advantage. The two-state model produces a single continuous signal (π_t^{filter}) that XGBoost can condition on in a straightforward way: higher values indicate more stress. With $K \geq 3$, the additional states may capture dimensions of the economy that are orthogonal to the continuation-reversal mechanism, for example a "growth" state characterised by strong fundamentals but moderate stress indicators. Such a state would conflate two economically distinct situations (a genuine expansion vs. a recovery from panic) and could introduce noise into the regime signal, degrading the model's ability to distinguish continuation from reversal. The two-state framework avoids this by restricting the regime classification to a single dimension, the degree of market stress, which is the dimension most relevant to momentum's cross-sectional structure.

G.5 January Exclusion

Momentum is known to reverse in January (Jegadeesh and Titman, 1993): past losers outperform past winners, partially offsetting momentum's annual profitability. To verify that XGB's performance is not driven by a January seasonal, Table 20 recomputes all metrics after excluding the 14 January months from the test period. XGB's Sharpe ratio declines marginally from 1.11 to 1.06, confirming that the regime-momentum interaction operates throughout the calendar year. Fixed 12-month momentum improves slightly when January is excluded (0.06 to 0.14), consistent with the well-documented January reversal in unconditional momentum.

	Ann. Ret	Ann. Vol	Sharpe	Months
Full sample (baseline)				
XGB	21.7%	19.5%	1.11	167
Fixed 12-mo mom	−1.9%	26.1%	0.06	167
Excluding January				
XGB	20.9%	19.9%	1.06	153
Fixed 12-mo mom	0.1%	25.6%	0.14	153

Table 20: January exclusion test. Performance is recomputed after dropping all 14 January months from the test period.

Appendix H: External Validity

H.1 Alternative Train/Test Splits

Table 21 reports the baseline train/test split performance. The 25-year expanding-window backtest (Section 5.3.2) provides further out-of-sample evidence across alternative training windows and prediction horizons.

Train / Test split	N_{test}	LR Sharpe	XGB Sharpe
1990–2010 / 2011–2025 (baseline)	167	−0.01	1.11

Table 21: Out-of-sample performance under the baseline train/test split (long-short, mom+π, 50-seed ensemble). LR collapses while XGB achieves a Sharpe of 1.11 with highly significant factor model alphas.

In the baseline split (1990–2010 training, 2011–2025 test), XGB achieves a Sharpe of 1.11 while LR collapses (−0.01), confirming the nonlinear regime-momentum interaction documented in Section 5.1.

H.2 25-Year Expanding-Window Out-of-Sample Methodology

The historical evidence reported in Section 5.3.2 comes from a separate analysis from the one above: a 25-year out-of-sample backtest in which the full pipeline (HMM + XGBoost) is re-estimated every January, with annual retraining initialised in 1995 and each year's predictions made by a model trained only on data available through the prior December. Each retraining uses the production seed configuration: 200 HMM seeds (Gibbs sampler with 2000 iterations, 500 burn-in), 50 XGBoost seeds (depth 4, learning rate 0.05, 500 trees), and the same long-short construction described in Appendix C.1.

The HMM's panic-state label is fixed by an economically motivated sign convention rather than the data-driven percentile comparison used elsewhere. The convention encodes the known direction of each feature in panic regimes: -1 for drawdown, $+1$ for cross-sectional dispersion, -1 for relative market participation, and $+1$ for the credit spread. The data-driven label-switching procedure used in the main pipeline is unstable on short training windows with few observed crisis months (under 20), so the fixed-sign fallback activates for retrainings before 2002. From 2002 onward the data-driven procedure resumes, by which point the training window contains the dot-com bust as a calibration anchor. The HMM seed loop is parallelised across CPU cores via `multiprocessing.Pool`; the parallel implementation is bit-identical to the serial reference (verified in the test suite by comparing per-seed pi-filter arrays).

Each retraining produces 12 monthly out-of-sample long-short returns, which are stitched into a single monthly return series; this thesis reports the 25-year sub-sample from January 2000 to November 2024 (299 months). Every prediction is strictly causal: the model used for month t saw no information after month $t-1$. The 167-month sub-window from January 2011 onward overlaps the main test period reported in Section 5.1 and serves as a methodological cross-check; expanding-window OOS produces a Sharpe of 0.88 over that window, within the production result's bootstrap interval.

H.3 Factor Model Regressions

To test whether XGB's out-of-sample return is explained by known risk factors, monthly returns are regressed on five standard factor models: CAPM, Fama–French three-factor, Carhart four-factor, Fama–French five-factor, and a six-factor specification that augments FF5 with the momentum factor (UMD). Newey–West standard errors (6 lags) are used throughout. Table 22 reports the intercept (alpha) and its t-statistic under each specification; Table 23 repeats the exercise for the fund-augmented variant. Alphas that survive all five factor models indicate return that cannot be attributed to market, size, value, profitability, investment, or momentum exposure.

Model	α (%)	$t(\alpha)$	Mkt-RF	SMB	HML	UMD	RMW	CMA
CAPM	23.4	4.08	-0.14					
FF3	23.2	4.48	-0.14	+0.04	-0.17			
Carhart	24.3	4.75	-0.19	-0.01	-0.22	-0.20		
FF5	22.9	4.62	-0.13	+0.08	-0.23		+0.06	+0.12
FF6	24.1	4.81	-0.18	+0.01	-0.31	-0.22	+0.01	+0.21

Table 22: Factor model regressions for XGB (mom+π). α is annualised. t-statistics use Newey–West standard errors (6 lags).

For completeness, Table 23 reports the same regressions for the fund-augmented variant (XGB: mom+π+fund) discussed in Section 5.1.3. Alphas remain statistically significant under all five specifications ($t > 2.9$) but are materially smaller than the baseline's, consistent with the dilution documented in the main text.

Model	α (%)	$t(\alpha)$	Mkt-RF	SMB	HML	UMD	RMW	CMA
CAPM	17.6	3.04	-0.16					
FF3	17.1	3.12	-0.14	-0.06	-0.13			
Carhart	18.2	3.22	-0.19	-0.12	-0.18	-0.20		
FF5	16.4	3.02	-0.11	+0.01	-0.30		+0.09	+0.35
FF6	17.7	3.17	-0.16	-0.06	-0.39	-0.23	+0.04	+0.44

Table 23: Factor model regressions for XGB (mom+π+fund). α is annualised. Newey–West t-statistics (6 lags).

H.4 Block Bootstrap Inference

To assess the sampling variability of XGB's Sharpe ratio and test its outperformance formally, a block bootstrap is applied to the monthly return series. Returns are resampled with replacement in 12-month blocks to preserve autocorrelation, producing 10,000 synthetic 167-month paths (fixed seed 42). The Sharpe ratio is recomputed on each resample to obtain a 95% confidence interval, and the same block draws are used for XGB and each benchmark to form a paired test of the Sharpe difference. The bottom panel repeats the exercise on XGB returns split by regime.

	Sharpe	95% CI	vs XGB (p)
XGB	1.11	$[0.67, 1.53]$	—
LR	-0.01	$[-0.41, 0.42]$	0.001***
DET (Formula)	0.11	$[-0.36, 0.59]$	0.008***
Fixed 12-mo mom	0.06	$[-0.43, 0.59]$	0.005***
Fixed 1-mo mom	0.26	$[-0.21, 0.73]$	0.021**
XGB regime-conditional Sharpe			
XGB Calm (n =108)	0.84	$[0.29, 1.32]$	—
XGB Panic (n =59)	1.54	$[0.85, 2.22]$	—
Panic − Calm	0.70	[-0.16, 1.59]	0.109

Table 24: Block bootstrap confidence intervals for Sharpe ratios and paired tests of XGB's outperformance. Uses 12-month blocks and 10,000 resamples with a fixed seed. The "vs XGB (p)" column reports two-sided p-values from paired tests where both strategies are resampled on the same dates. $^{*}\,p < 0.10$; $^{**}\,p < 0.05$; $^{***}\,p < 0.01$.

XGB's Sharpe confidence interval is [0.67, 1.53], comfortably above zero, while every benchmark's interval straddles zero. The paired tests reject equality of Sharpe ratios between XGB and each benchmark at $p < 0.025$. Within XGB, the panic-period Sharpe exceeds the calm-period Sharpe in point estimate (1.54 vs 0.84), but with only 59 panic months the paired difference is not statistically significant ($p = 0.109$); the 95% interval $[-0.16, 1.59]$ straddles zero. The point-estimate gap is consistent with the regime-conditional selection mechanism described in Section 5.2, but its statistical confirmation would require a longer panic sample. These intervals describe sampling variability within the 2011–2025 sample; out-of-sample generalisation is addressed separately by the 25-year expanding-window backtest (Section 5.3.2) and the sub-period analysis (Appendix G.1).

H.5 Leg-Level CAPM Betas by Regime

The D&M comparison in Section 5.3.3 characterises XGB's contribution as stock selection rather than exposure scaling. A direct test of this framing is to compute leg-level CAPM betas separately by regime: if XGB reorganises which stocks it holds across regimes (cross-sectional re-ranking), the long-leg and short-leg betas should shift between calm and panic in opposite directions; if instead XGB mechanically scales exposure, both legs' betas should move together.

Table 25: Leg-level CAPM betas by regime, 2011–2025 OOS test period. Panic months are those with $\pi_t^{\text{filter}} \geq 0.5$. Newey–West standard errors (3 lags) in parentheses.

Strategy	Leg	Full	Calm	Panic
Fixed 12-mo mom	Long	1.09 (0.06)	1.09 (0.09)	1.07 (0.08)
Fixed 12-mo mom	Short	1.74 (0.11)	1.80 (0.21)	1.69 (0.13)
Method 2: XGB	Long	1.55 (0.08)	1.40 (0.11)	1.65 (0.10)
Method 2: XGB	Short	1.10 (0.06)	1.22 (0.09)	0.98 (0.05)

The pattern in Table 25 is consistent with the cross-sectional re-ranking interpretation. XGB's long-leg beta rises from 1.40 in calm to 1.65 in panic (*tilt amplification*: in panic, the long leg holds names with higher market sensitivity), while the short-leg beta falls from 1.22 to 0.98. Fixed 12-month momentum, by contrast, holds both leg-level betas roughly stable across regimes (long: 1.09 calm, 1.07 panic; short: 1.80 calm, 1.69 panic), consistent with the unconditional rank-based construction. The difference is direct evidence that the regime signal acts on *which* stocks are held rather than on *how much*.

H.6 Per-cluster Feature Signature ($K = 4$ Partition)

Section 5.2.2 reports a $K = 4$ partition of the 167 test months and characterises each cluster narratively. Table 26 provides the full per-cluster feature signature for completeness: cross-section context features (regime composition, momentum landscape, dispersion, skewness, recent strategy performance) computed at the month level and averaged across each cluster's months; long-leg pick raw momentum at each horizon, averaged across each cluster's months; and long-leg pick cross-sectional z-score (deviation from the cross-section mean at each horizon, in units of cross-section std) averaged across cluster months. Per-cluster Sharpes and other strategy outcomes appear in Table 8 in the main text.

Feature	Cluster 1 *calm cont.*	Cluster 2 *mild cont.*	Cluster 3 *post-panic*	Cluster 4 *deep crisis*
Cross-section context				
Cross-section mom (overall mean)	13.8%	7.3%	13.5%	−9.3%
Cross-section mom (short tertile, $h = 1$–4)	6.1%	4.4%	3.0%	−7.1%
Cross-section mom (mid tertile, $h = 5$–8)	14.6%	7.6%	12.4%	−11.1%
Cross-section mom (long tertile, $h = 9$–12)	20.8%	9.7%	25.0%	−9.6%
Cross-section dispersion (short)	0.23	0.27	0.28	0.23
Cross-section dispersion (mid)	0.44	0.46	0.56	0.35
Cross-section dispersion (long)	0.60	0.59	0.85	0.45
Cross-section skewness (short)	4.6	7.2	6.4	3.7
Cross-section skewness (mid)	6.2	7.6	6.2	5.6
Cross-section skewness (long)	6.7	8.1	6.5	5.7
π^{filter} frequency, prior 6 months	0.27	0.37	0.30	0.51
π^{filter} frequency, prior 12 months	0.31	0.32	0.41	0.30
Strategy Sharpe, prior 12 months	1.22	1.04	1.15	1.08
Long-leg pick raw momentum by horizon				
Long-leg pick mom, $h = 1$	7.1%	3.1%	−3.4%	−15.3%
Long-leg pick mom, $h = 2$	13.5%	8.6%	−5.5%	−21.6%
Long-leg pick mom, $h = 3$	18.3%	8.8%	−5.3%	−26.3%
Long-leg pick mom, $h = 4$	23.1%	10.6%	−5.8%	−30.8%
Long-leg pick mom, $h = 5$	30.8%	13.8%	−5.9%	−35.0%
Long-leg pick mom, $h = 6$	40.2%	18.0%	−3.2%	−38.3%
Long-leg pick mom, $h = 7$	49.4%	20.7%	−1.3%	−42.1%
Long-leg pick mom, $h = 8$	59.2%	23.7%	0.0%	−44.9%
Long-leg pick mom, $h = 9$	61.3%	23.1%	1.1%	−46.5%
Long-leg pick mom, $h = 10$	60.7%	21.9%	1.3%	−47.2%
Long-leg pick mom, $h = 11$	61.1%	22.8%	1.2%	−47.3%
Long-leg pick mom, $h = 12$	57.9%	18.5%	−5.3%	−47.5%
Long-leg pick cross-sectional z-score by horizon				
Long-leg z-curve, $h = 1$	0.26	0.04	−0.28	−0.73
Long-leg z-curve, $h = 2$	0.38	0.14	−0.31	−0.74
Long-leg z-curve, $h = 3$	0.40	0.11	−0.34	−0.75
Long-leg z-curve, $h = 4$	0.44	0.10	−0.36	−0.80
Long-leg z-curve, $h = 5$	0.54	0.16	−0.35	−0.85
Long-leg z-curve, $h = 6$	0.65	0.23	−0.29	−0.83
Long-leg z-curve, $h = 7$	0.73	0.27	−0.26	−0.85
Long-leg z-curve, $h = 8$	0.80	0.32	−0.22	−0.87
Long-leg z-curve, $h = 9$	0.76	0.29	−0.23	−0.88
Long-leg z-curve, $h = 10$	0.72	0.24	−0.25	−0.87
Long-leg z-curve, $h = 11$	0.67	0.22	−0.26	−0.86
Long-leg z-curve, $h = 12$	0.54	0.12	−0.35	−0.87

Table 26: Per-cluster feature signature for the $K = 4$ partition, 2011–2024 test period. Features are grouped into three blocks: (1) cross-section context features (regime composition, momentum landscape, dispersion, skewness, recent strategy performance) computed at the month level and averaged across each clusterś months; (2) long-leg pick raw momentum at each horizon, averaged across each clusterś months; (3) long-leg pick cross-sectional z-score (deviation from the cross-section mean at each horizon, in units of cross-section std) averaged across cluster months. Per-cluster Sharpes and other strategy outcomes appear in Table 8; this table is the full descriptive feature signature.

H.7 Cluster-Level Cross-Validation

Three diagnostics extend the regime-aggregate findings of Section 5.2 across the $K = 4$ partition.

Leg-betas across the $K = 4$ partition

XGB retains long-leg $\beta >$ short-leg β in every cluster (spreads +0.68, +0.40, +0.49, +0.43 for clusters 1–4), while unconditional 12-month momentum exhibits leg-beta inversion in every cluster (−0.24, −0.87, −0.75, −0.42; Table 27). Daniel and Moskowitz (2016)'s crash mechanism is latent across every cluster type; XGB escapes it everywhere through compositional re-ranking.

		XGB			Fixed 12-mo mom		
Cluster	n	β_L	β_S	Spread	β_L	β_S	Spread
1 (calm continuation)	53	1.65 (0.22)	0.97 (0.13)	0.68	1.32 (0.22)	1.57 (0.24)	−0.24
2 (mild continuation)	62	1.60 (0.15)	1.20 (0.09)	0.40	1.04 (0.08)	1.91 (0.17)	−0.87
3 (post-panic recovery)	31	1.46 (0.16)	0.97 (0.11)	0.49	1.09 (0.17)	1.84 (0.22)	−0.75
4 (deep crisis)	21	1.49 (0.08)	1.06 (0.09)	0.43	1.07 (0.10)	1.49 (0.14)	−0.42

Table 27: Leg-level CAPM betas by K=4 cluster, 2011–2024 test period. XGB (regime-conditional XGBoost ensemble) versus unconditional 12-month momentum, side-by-side. Standard errors in parentheses. Spread = $\beta_L - \beta_S$.

π_t^{filter}'s SHAP share by cluster

Combined long-and-short SHAP shares are 0.50, 0.46, 0.54, 0.24 for clusters 1–4 (Table 28). The pattern is consistent with the routing interpretation: π_t^{filter} contributes most where the cross-section is ambiguous (cluster 3) and least where it is unambiguously crisis-shaped (cluster 4, where momentum splits alone suffice to produce the loser portfolio).

Cluster	n	Long π share	Short π share	Combined π share
1 (calm continuation)	53	34.0%	50.1%	49.9%
2 (mild continuation)	62	26.9%	46.5%	46.0%
3 (post-panic recovery)	31	27.8%	53.7%	53.6%
4 (deep crisis)	21	11.4%	38.1%	24.4%

Table 28: π^{filter}'s SHAP share by K=4 cluster, 2011–2024 test period. Long and short shares are computed over long-leg and short-leg picks respectively; combined is over all stock-months in the cluster.

Short-leg structure (future work)

The $K = 4$ partition was defined on long-leg z-curves; applying the same labels to the short leg shows the short basket is structurally distinct from a negated long-leg, with short-vs-negated-long correlations of +0.94, +0.89, +0.19, and −0.84 for clusters 1–4 (Table 29). The month-1 short-leg spike in cluster 4 (+0.57) is consistent with Nagel

(2012)'s finding that short-horizon reversal returns spike during high-volatility periods: month 1 carries 10.6% of panic short-leg momentum SHAP versus 5.3% in calm. A full short-leg analysis is left to future work.

Cluster	n	z_1	$\bar{z}$	Basket	$\rho(-z^L)$
1 (calm continuation)	53	−0.062	−0.365	391	0.943
2 (mild continuation)	62	0.101	−0.174	334	0.888
3 (post-panic recovery)	31	0.209	0.241	382	0.190
4 (deep crisis)	21	0.568	0.170	353	−0.835

Table 29: Short-leg z-curve summary by K=4 cluster, 2011–2024 test period. z_1 is the cluster-mean cross-sectional z-score of short-leg picks at the 1-month horizon; $\bar{z}$ averages across all 12 horizons. Basket size is mean stocks per month. $\rho(-z^L)$ is the correlation between the short-leg z-curve centroid and the negated long-leg z-curve centroid; values below 0.95 indicate structural asymmetry.

H.8 Emission Distribution Robustness

To verify that regime identification is robust to the choice of emission distribution, the HMM is re-estimated with multivariate Student-t emissions across a grid of degrees of freedom $\nu \in \{3, 5, 7, 10, 15, 20, 30, 50, 100\}$.

The Student-t model is estimated via data augmentation (Geweke, 1993): each observation receives a latent weight $w_t \sim \text{Gamma}(\nu/2, \nu/2)$, so that $\mathbf{z}_t \mid s_t{=}k, w_t \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k/w_t)$. Marginalising over w_t yields a multivariate t with ν degrees of freedom. Crucially, conditional on w_t the emission is Gaussian, so the Normal-Inverse-Wishart conjugate updates carry over with weighted sufficient statistics, preserving the efficiency of the Gibbs sampler.

ν	LL (train)	LL (test)	BIC	Agreement	Corr(π)
3	−882.7	−701.1	1929.9	91.4%	0.872
5	−829.0	−669.8	1822.5	96.8%	0.960
7	−838.3	−698.2	1841.2	98.0%	0.984
10	−815.8	−644.3	1796.1	96.8%	0.960
15	−800.1	−651.0	1764.8	98.3%	0.997
20	−802.9	−621.7	1770.4	99.3%	0.997
30	−802.6	−640.6	1769.8	98.5%	0.993
50	−796.6	−658.1	1757.7	98.8%	0.996
100	−792.3	−643.5	1749.0	98.8%	0.987
Normal	−787.5	−642.5	**1739.5**	—	—

Table 30: Student-t vs. Normal emission HMM comparison. LL = log-likelihood; Agreement = fraction of months with identical binary regime classification as the Normal HMM; Corr(π) = correlation of filtered panic probabilities.

The Normal model achieves the lowest BIC (2257.3 vs. 2257.6 for the best Student-t at

$\nu = 100$), indicating no meaningful improvement from heavier tails. Regime classifications agree with the Normal HMM on at least 97.8% of months across all ν values, and the correlation of filtered panic probabilities exceeds 0.98 everywhere. These results confirm that the Gaussian emission specification is robust for this application.

Appendix I: Label Switching: Sign Correction Procedure

Because the four HMM features respond to stress in different directions, the expected direction of each feature during stress is determined empirically. For each feature j, the 95th percentile is computed separately on training months falling within known crisis windows (the dot-com bust of 2000–2002 and the Global Financial Crisis of 2007–2009) and on all remaining training months:

$$\text{sign}_j = \begin{cases} +1 & \text{if } q_{0.95}^{\text{crisis}}(z_j) \geq q_{0.95}^{\text{normal}}(z_j), \\ -1 & \text{otherwise.} \end{cases} \tag{41}$$

Features that spike upward during crises receive $\text{sign}_j = +1$; features that decline receive $\text{sign}_j = -1$. The panic state is then identified as the regime $k \in \{0, 1\}$ whose posterior mean vector $\hat{\boldsymbol{\mu}}_k$ best aligns with the crisis direction:

$$k^* = \arg \max_{k \in \{0,1\}} \sum_{j=1}^{D} \text{sign}_j \cdot \hat{\mu}_{k,j} \tag{42}$$

The state labelled k^* is designated as "panic" and the other as "calm."

Appendix J: Convergence Diagnostics

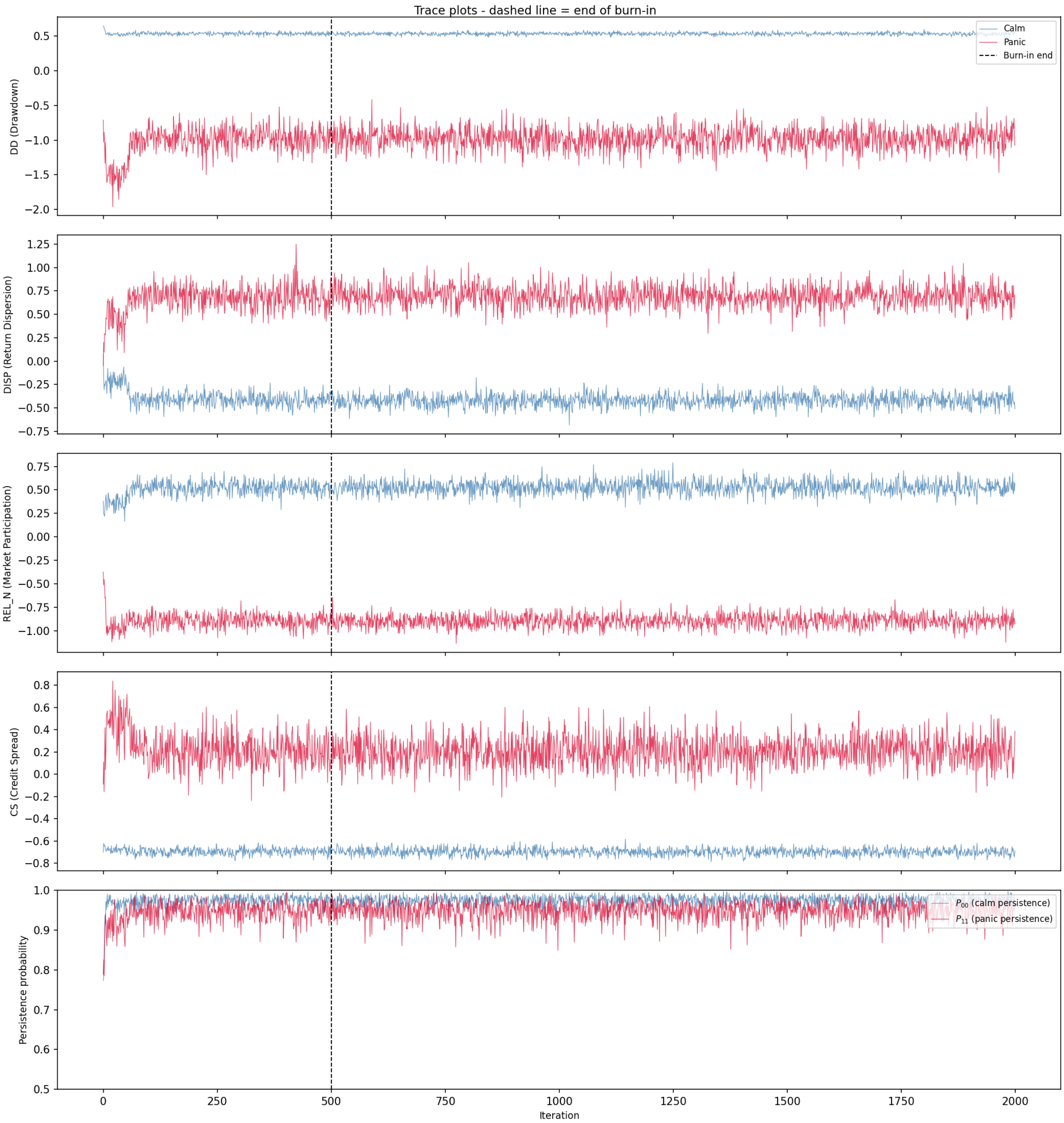

Figure 13: Trace plots of the Gibbs sampler. Each panel shows sampled regime means (calm in blue, panic in red) for one HMM feature. The dashed line marks the end of burn-in.

J.1 Gelman–Rubin Diagnostic

The Gelman–Rubin $\hat{R}$ statistic (Gelman and Rubin, 1992; Gelman et al., 2013) assesses convergence by comparing within-chain and between-chain variance across the five independent Gibbs sampler chains. For each parameter θ, $\hat{R} = \sqrt{\hat{V}/W}$, where W is the mean within-chain variance and $\hat{V} = \frac{n-1}{n} W + \frac{B}{n}$ combines within-chain variance with the between-chain variance B. Values near 1.0 indicate that the chains have converged to the same posterior distribution; $\hat{R} < 1.1$ is the standard threshold.

Parameter	$\hat{R}$	Status
$\mu_{\text{panic}}(\text{DD})$	1.000	OK
$\mu_{\text{panic}}(\text{DISP})$	1.000	OK
$\mu_{\text{panic}}(\text{REL\_N})$	1.000	OK
$\mu_{\text{panic}}(\text{CS})$	1.000	OK
$\mu_{\text{calm}}(\text{DD})$	1.000	OK
$\mu_{\text{calm}}(\text{DISP})$	1.001	OK
$\mu_{\text{calm}}(\text{REL\_N})$	1.000	OK
$\mu_{\text{calm}}(\text{CS})$	1.000	OK
P_{00} (calm persistence)	1.000	OK
P_{01} (calm $\to$ panic)	1.000	OK
P_{10} (panic $\to$ calm)	1.000	OK
P_{11} (panic persistence)	1.000	OK

Table 31: Gelman–Rubin $\hat{R}$ statistics for all HMM parameters, computed across 5 independent chains (1,500 post-burn-in draws each). All values are below 1.01, well within the convergence threshold of 1.1.

J.2 Seed Convergence of the Regime Signal

Gelman–Rubin and the trace plots assess whether a single Gibbs chain has converged to its stationary posterior, but they do not quantify how many chains are needed to stabilise the *downstream* Sharpe ratio. The regime signal π_t^{filter} enters the pipeline as the average of N_s seed-specific forward filters (Section 3.1.5), and because XGBoost is a nonlinear function of π_t^{filter}, MCMC noise in any individual chain does not cancel linearly.

To measure this dependence, we subsample $k \in \{1, 2, 3, 5, 10, 15, 20, 30, 50, 75, 100\}$ chains without replacement from the seed pool, average the forward filter over those k chains, retrain the XGBoost ensemble on the resulting signal, and record the out-of-sample Sharpe. For each $k < 100$ we repeat on 30 random subsets; $k = 100$ exhausts the pool.

k	n_{sub}	Mean	Median	Std	P25	P75	Min	Max
1	30	1.040	1.031	0.078	0.985	1.090	0.915	1.205
2	30	1.062	1.081	0.069	1.019	1.103	0.894	1.217
3	30	1.049	1.044	0.043	1.014	1.075	0.973	1.143
5	30	1.042	1.042	0.047	1.002	1.083	0.967	1.154
10	30	1.053	1.050	0.036	1.038	1.066	0.965	1.145
15	30	1.075	1.079	0.029	1.055	1.097	1.008	1.139
20	30	1.078	1.080	0.022	1.065	1.090	1.025	1.122
30	30	1.089	1.082	0.025	1.069	1.109	1.052	1.137
50	30	1.093	1.092	0.015	1.082	1.101	1.067	1.124
75	30	1.098	1.102	0.012	1.091	1.105	1.067	1.118
100	1	1.118	1.118	0.000	1.118	1.118	1.118	1.118

Table 32: XGBoost ensemble seed convergence. For each subset size k, we draw n_{sub} random subsets of k seeds from the full 100-seed pool, average their predictions, and report the resulting out-of-sample L/S Sharpe ratio. Mean Sharpe stabilizes by $k \approx 50$, justifying the production choice; standard deviation across subsets shrinks as $1/\sqrt{k}$.

Two patterns emerge. First, the mean Sharpe rises from 1.04 at $k = 1$ to 1.12 at $k = 100$, so a single-chain point estimate understates the strategy's performance by roughly 0.08 in Sharpe units. The gap closes monotonically, with most of the lift realised by $k = 30$. Second, the interquartile range across random subsets collapses from 0.10 at $k = 1$ to 0.02 at $k = 50$: once $k \geq 50$, any two random subsets of the seed pool produce nearly the same XGBoost portfolio. The production configuration ($N_s = 200$) sits well inside this plateau, which is why the 1.11 Sharpe reported in Section 5 is insensitive to which specific seeds were drawn. The practical implication is methodological rather than cosmetic: a single-chain HMM would not reliably support the main conclusions, so the seed ensemble is a requirement of the design, not a late-stage performance tuning.

Appendix K: International HMM Feature Selection

For the international robustness check (Section 5.3.4), the HMM feature set is selected per region using the same 4-pass procedure as the US production specification (Section 4.2): a quality screen on MCMC convergence and crisis-window alignment, a portfolio-Sharpe ranking on the surviving combinations, a definitive validation at higher seed counts, and an additional stability check across random seed partitions. The international candidate pool comprises the five locally-available stress features (DD, VOL, DISP, REL_N, BANK_REL); BANK_REL is the value-weighted bank-sector excess return over the broad regional market, sign-flipped so high values indicate stress, and substitutes for the US-specific Moody's BAA-AAA spread. With DD required, the procedure evaluates fifteen combinations of size one to four. Both UK and Japan select *DD+VOL+REL_N* as the regional feature set. The transplanted-from-US template (DD+DISP+REL_N+BANK_REL) fails the Pass-1 quality screen for both regions (minimum effective sample size of 14 and 4 respectively), motivating per-region selection. Pass-by-pass CSV results and chosen-features JSON manifests are saved under `results/thesis/` with the prefix `intl_<region>_hmm`.

Appendix L: Fundamental Feature Definitions

Ten firm-level characteristics are computed from Compustat quarterly data (`fundq`), linked to CRSP via the CCM bridge table. All fundamentals are lagged four months from fiscal quarter end to ensure real-time availability. Cash-flow items (`oancfy`, `capxy`) are reported year-to-date and are differenced within each fiscal year to recover the per-quarter contribution before the trailing twelve-month sum is taken.

- **Book-to-market (bm):** Book equity (`ceqq`) divided by market equity.
- **Return on equity (roe):** Income before extraordinary items (`ibq`) divided by book equity.
- **Gross profitability:** Revenue minus cost of goods sold, scaled by total assets: $(\texttt{revtq} - \texttt{cogsq})/\texttt{atq}$.
- **Asset growth:** Year-over-year growth in total assets: $\texttt{atq}_t/\texttt{atq}_{t-4} - 1$.
- **Leverage:** Total debt divided by total assets: $(\texttt{dlttq} + \texttt{dlcq})/\texttt{atq}$.
- **Earnings growth:** Year-over-year growth in earnings, transformed as $\text{sign}(g) \cdot \log(1 + |g|)$.
- **Log market equity:** Natural logarithm of market capitalisation (price $\times$ shares outstanding), lagged one month.
- **Operating cash flow over assets (cfo_a):** Trailing twelve-month operating cash flow (`oancfy`, converted to quarterly and summed) scaled by total assets.
- **Free cash flow over assets (fcf_a):** Trailing twelve-month operating cash flow minus trailing twelve-month capital expenditure (`capxy`), scaled by total assets.
- **Accruals:** The accrual component of earnings, $(\texttt{ibq}_{\text{TTM}} - \texttt{oancf}_{\text{TTM}})/\texttt{atq}$, which captures the non-cash portion of reported earnings.

Fundamentals are winsorised at the 1st and 99th percentiles (5th and 95th for asset growth) using training-sample bounds. The cash-flow features (cfo_a, fcf_a, accruals) are available for roughly 83% of stock-months; the remaining seven characteristics exceed 82% coverage individually, and XGBoost handles any residual missingness natively via its default-branch logic.